Tactical asset allocation for EuroLife Target 2035

Asset/Risk Management II

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Task 1

Data

All data are monthly. The main data set consists of month-end values from Bloomberg, spanning from November 1996 to September 2026. The data consists of the following for each part of the question:

Asset classes. We have the total return indices (all income reinvested) for the seven asset classes of the SAA:

#	Asset class	Index	Currency
1	US equity	MSCI USA Net Total Return	USD
2	Euro area equity	MSCI EMU Net Total Return	EUR
3	Developed markets equity ex US and ex euro area	built from MSCI World ex USA and MSCI EMU Net TR, with market caps	USD
4	Emerging markets equity	MSCI Emerging Markets Net Total Return	USD
5	Euro government bonds	Bloomberg Euro Aggregate Treasury Total Return	EUR
6	US Treasuries	Bloomberg US Treasury Total Return	USD
7	Euro investment-grade corporate bonds	Bloomberg Euro Aggregate Corporate Total Return	EUR

Cash rates and exchange rate. 3-month Euribor (the euro cash rate), the 3-month USD money market rate (the USD cash rate: USD LIBOR until it ended in September 2024, then 3-month Term SOFR plus the official fallback spread of 0.26 percentage points) and the EUR/USD spot rate, quoted in USD per EUR.

Predictors for the views. MSCI price indices and estimated dividend yields for the four equity regions, and the trailing dividend yield of MSCI USA; yields to worst and durations of the bond indices; German and US 10-year government bond yields and the 3-month US Treasury bill yield; option-adjusted spreads of euro corporate bonds; euro area HICP inflation; and the 5y5y euro inflation swap rate.

Hedge funds (task 7c). The Credit Suisse Hedge Fund Index (Bloomberg: HEDGNAV), net of fees and quoted in USD, as a proxy for the hedge fund allocation the board is considering.

Long histories for the predictive regressions. Three free sources extend some series much further back: Robert Shiller's US stock market data (S&P Composite prices, dividends, earnings and the cyclically adjusted P/E ratio CAPE, monthly from 1871), the Deutsche Bundesbank's zero-coupon yields of German federal securities with 1 to 5 years to maturity (month-end, from 1972), and the monthly return of 1-month US Treasury bills from Kenneth French's data library (from July 1926), the US short rate for the long equity regression.

```r
data_file <- "Data/ARM_data_clean.xlsx"

panel <- as.data.frame(readxl::read_excel(data_file, sheet = "data"))   # one row per month, Nov 1996 to
↪  Sep 2026
panel$date <- as.Date(panel$date)
long_history <- as.data.frame(readxl::read_excel(data_file, sheet = "long_history", guess_max = 1e5))   #
↪  from 1871
long_history$date <- as.Date(long_history$date)
info <- as.data.frame(readxl::read_excel(data_file, sheet = "info"))     # one row per series

# the P/E ratios of the clean file are not used by any view
unused_series <- c("pe_us", "pe_emu", "pe_em", "pe_wxus")
panel <- panel[, setdiff(names(panel), unused_series)]
info  <- info[!info$name %in% unused_series, ]

# hedge fund index (task 7c) from its own file, matched by month since its dates are last business days
hf_index <- as.data.frame(readxl::read_excel("HF_Index.xlsx"))
panel$hf_hedgnav_usd <- hf_index[[2]][match(format(panel$date, "%Y-%m"), format(as.Date(hf_index$Date),
↪  "%Y-%m"))]

hf_info <- info[1, ]
hf_info[c("name", "description", "source", "currency", "unit", "group", "role")] <-
  list("hf_hedgnav_usd", "Credit Suisse Hedge Fund Index (HEDGNAV)", "Bloomberg", "USD", "index", "Hedge
  ↪  funds", "Hedge fund proxy (task 7c)")
is_long_history <- info$group == "Long history"
info <- rbind(info[!is_long_history, ], hf_info, info[is_long_history, ])   # the long histories stay
↪  last
```

How the USD cash rate is built

The case asks for 3-month money market rates. For the euro this is 3-month Euribor, the rate at which banks lend to each other unsecured. The matching USD rate is 3-month USD LIBOR, but LIBOR was phased out: the panel of banks stopped submitting in June 2023, and until September 2024 LIBOR was published only in a "synthetic" form, defined as 3-month Term SOFR plus a fixed spread of 0.26161 percentage points. Term SOFR is the market's forward-looking 3-month rate derived from SOFR, the overnight rate on repurchase agreements secured by Treasuries. Because it is secured, it is lower than an unsecured bank rate, and the fixed spread (set by ISDA as the median LIBOR-SOFR difference over the previous five years) makes up for this.

We therefore use 3-month USD LIBOR up to its last value in September 2024 and 3-month Term SOFR plus 0.26161 percentage points from October 2024. Since the synthetic LIBOR of July 2023 to September 2024 is already defined this way, there is no jump at the splice. The table shows the rates around the transition. The 3-month Treasury bill yield is shown for comparison: it is a government rate and usually lower, and it would understate the interest rate differential in x_t.

Table 2: USD 3-month rates around the end of LIBOR, in percent per year

Month-end	3M USD LIBOR	3M Term SOFR	Term SOFR + 0.26161	USD cash rate used	3M T-bill
Dec 2008	1.43			1.43	0.08
Jan 2019	2.74	2.45	2.72	2.74	2.38
Jun 2023	5.55	5.27	5.53	5.55	5.28
Jul 2023	5.63	5.37	5.63	5.63	5.40
Sep 2024	4.85	4.59	4.85	4.85	4.62
Oct 2024		4.56	4.82	4.82	4.54
Sep 2026		4.08	4.34	4.34	4.11

Data Cleaning

Keeping only months where every column is available would leave only January 2019 to September 2024: several series start late (the estimated dividend yields in 2005/06, 3-month Term SOFR in 2019), and USD LIBOR ends in September 2024. This would throw away most of the data, which is crucial, for example, when we compute the covariance matrix. Therefore, we require complete data only for the series that build the returns, not for the predictors. This results in two separate datasets:

- `core`: every series needed to compute the asset class returns (the asset indices, the two cash rates and EUR/USD)
- `panel`: everything, including predictors with their own (longer or shorter) histories.

```
# series needed to compute the asset class returns: asset indices, cash rates and EUR/USD
return_series <- info$name[info$group %in% c("Asset class", "Cash rate and FX")]

# keep only months where all of them are available
core <- panel[complete.cases(panel[, return_series]), c("date", return_series)]

# check: no month is missing between the first and last month of core
all_months <- seq(min(core$date) + 1, max(core$date) + 1, by = "month") - 1
stopifnot(nrow(core) == length(all_months))
```

`core` runs from Dec 1998 to Sep 2026 (334 month-ends, so 333 monthly returns). The start is set by 3-month Euribor, the euro-denominated MSCI EMU index and MSCI EM, which all begin in December 1998.

Documentation

The case asks us to state each series' name, source and quotation currency. All series come from Bloomberg except the long histories, whose source is given in the description. The table also shows what each series is used for, its sample period, the number of observations and any interior gaps (months missing after the series has started).

```
coverage <- function(df, vars) {
  do.call(rbind, lapply(vars, function(v) {
    ok <- which(!is.na(df[[v]]))
    span <- df[min(ok):max(ok), v]
    data.frame(start = format(df$date[min(ok)], "%Y-%m"), end = format(df$date[max(ok)], "%Y-%m"),
               n = length(ok), gaps = sum(is.na(span)))
  }))
}
is_long <- info$group == "Long history"
series_info <- cbind(info[, c("name", "description", "source", "currency", "unit", "group", "role")],
                     rbind(coverage(panel, info$name[!is_long]), coverage(long_history,
                     ↪  info$name[is_long])))
```

Table 3: Data used in the case study (monthly)

Variable	Series	Currency	Used for	Sample	Obs.	Gaps
Asset class						
eq_us_usd	MSCI USA Net Total Return Index	USD	US equity	1996-11 to 2026-09	359	0
eq_emu_eur	MSCI EMU Net Total Return Index (in EUR)	EUR	Euro area equity	1998-12 to 2026-09	334	0
eq_wxus_usd	MSCI World ex USA Net Total Return Index (NDDUWXUS)	USD	Dev. ex US ex euro area equity (component)	1996-11 to 2026-09	359	0
eq_emu_usd	MSCI EMU Net Total Return Index in USD (NDDUEMU)	USD	Dev. ex US ex euro area equity (component)	1996-11 to 2026-09	359	0
mcap_emu	Market cap of MSCI EMU	USD	Dev. ex US ex euro area equity (weight)	1996-11 to 2026-09	359	0

Table 3: Data used in the case study (monthly) *(continued)*

Variable	Series	Currency	Used for	Sample	Obs.	Gaps
mcap_wxus	Market cap of MSCI World ex USA	USD	Dev. ex US ex euro area equity (weight)	1996-11 to 2026-09	359	0
eq_em_usd	MSCI Emerging Markets Net Total Return Index (NDUEEGF)	USD	Emerging markets equity	1998-12 to 2026-09	334	0
bd_egov_eur	Bloomberg Euro Aggregate Treasury Total Return (LEATTREU)	EUR	Euro government bonds	1998-06 to 2026-09	340	0
bd_ust_usd	Bloomberg US Treasury Total Return Unhedged (LUATTRUU)	USD	US Treasuries	1996-11 to 2026-09	359	0
bd_ecorp_eur	Bloomberg Euro Aggregate Corporate Total Return (LECPTREU)	EUR	Euro IG corporate bonds	1998-06 to 2026-09	340	0
Cash rate and FX						
euribor3m	3-month Euribor (EUR003M)	EUR	Euro cash rate	1998-12 to 2026-09	334	0
eurusd	EUR/USD spot rate, USD per EUR (EURUSD)	USD per EUR	Exchange rate	1996-11 to 2026-09	359	0
usd3m	USD 3-month money market rate: LIBOR to Sep 2024, then Term SOFR + 0.26161	USD	USD cash rate	1996-11 to 2026-09	359	0
USD rate components						
usd_libor3m	3-month USD LIBOR (synthetic from July 2023, ended September 2024)	USD	USD cash rate (to Sep 2024)	1996-11 to 2024-09	335	0
usd_tsofr3m	3-month CME Term SOFR	USD	USD cash rate (from Oct 2024, + 0.26161)	2019-01 to 2026-09	93	0
Predictor						
tbill3m	US 3-month Treasury bill yield (USGG3M)	USD	US short rate	1996-11 to 2026-09	359	0
px_us	MSCI USA Price Index (MXUS)	USD	US equity (dividends)	1996-11 to 2026-09	359	0
px_emu	MSCI EMU Price Index (MXEM; EMU, not EM)	EUR	Euro area equity (dividends)	1996-11 to 2026-09	359	0
px_em	MSCI Emerging Markets Price Index (MXEF)	USD	Emerging markets equity (dividends)	1996-11 to 2026-09	359	0
px_wxus	MSCI World ex USA Price Index (MXWOU)	USD	Dev. ex US equity (dividends)	1996-11 to 2026-09	359	0
dy_us	Estimated dividend yield, MSCI USA	-	US equity valuation	2006-01 to 2026-09	249	0
dy_emu	Estimated dividend yield, MSCI EMU	-	Euro area equity valuation	2006-03 to 2026-09	247	0
dy_em	Estimated dividend yield, MSCI Emerging Markets	-	Emerging markets equity valuation	2005-04 to 2026-09	258	0
dy_wxus	Estimated dividend yield, MSCI World ex USA	-	Dev. ex US equity valuation	2006-02 to 2026-09	248	0
y10_de	German 10-year government bond yield (GDBR10)	EUR	Euro government bond yields	1996-11 to 2026-09	359	0
y10_us	US 10-year Treasury yield (USGG10YR)	USD	US Treasury yields	1996-11 to 2026-09	359	0
oas_ecorp	Option-adjusted spread, Euro Aggregate Corporate (LECPOAS)	EUR	Euro corporate credit spread	1998-11 to 2026-09	317	18
hicp	Euro area HICP, all items (CPTFEMU)	EUR	Euro inflation	1996-11 to 2026-09	359	0
infl_5y5y	EUR 5y5y forward inflation swap rate (FWISEU55)	EUR	Euro inflation expectations	2004-04 to 2026-09	270	0
dy_us_trailing	Trailing 12-month gross dividend yield, MSCI USA	-	US equity valuation	1996-11 to 2026-09	359	0
ytw_egov	Yield to worst, Bloomberg Euro Aggregate Treasury	EUR	Euro government bond yields	1998-06 to 2026-09	340	0
ytw_ecorp	Yield to worst, Bloomberg Euro Aggregate Corporate	EUR	Euro corporate bond yields	1998-06 to 2026-09	340	0
oas_ecorp_swap	Option-adjusted spread to swaps, Euro Aggregate Corporate	EUR	Euro corporate credit spread	1999-01 to 2026-09	333	0
infl_yoy	Euro area HICP inflation, year on year (ECCPEMUY)	EUR	Euro inflation	1996-12 to 2026-09	358	0
dur_egov	Option-adjusted duration, Euro Aggregate Treasury	-	Euro government bond duration	1998-06 to 2026-09	340	0
dur_ust	Option-adjusted duration, US Treasury	-	US Treasury duration	1997-07 to 2026-09	351	0
dur_ecorp	Option-adjusted duration, Euro Aggregate Corporate	-	Euro corporate bond duration	1998-06 to 2026-09	340	0
Hedge funds						
hf_hedgnav_usd	Credit Suisse Hedge Fund Index (HEDGNAV)	USD	Hedge fund proxy (task 7c)	1996-11 to 2026-09	359	0
Long history						
sp_price	S&P Composite price, monthly average of daily closes (Shiller)	USD	US equity (dividends)	1871-01 to 2026-09	1869	0
sp_dividend	S&P Composite dividends, trailing 12 months (Shiller)	USD	US equity (dividends)	1871-01 to 2026-06	1866	0

Table 3: Data used in the case study (monthly) *(continued)*

Variable	Series	Currency	Used for	Sample	Obs.	Gaps
sp_earnings	S&P Composite earnings, trailing 12 months (Shiller)	USD	US equity valuation	1871-01 to 2026-06	1866	0
us_cpi	US consumer price index (Shiller)	-	US inflation	1871-01 to 2026-09	1869	0
cape	Cyclically adjusted P/E ratio: price over 10-year average real earnings (Shiller)	-	US equity valuation	1881-01 to 2026-09	1749	0
zero_de_1y	German zero-coupon yield, 1 year, month-end (Bundesbank, Svensson method)	EUR	Euro government bond yields	1972-09 to 2026-09	649	0
zero_de_2y	German zero-coupon yield, 2 years, month-end (Bundesbank, Svensson method)	EUR	Euro government bond yields	1972-09 to 2026-09	649	0
zero_de_3y	German zero-coupon yield, 3 years, month-end (Bundesbank, Svensson method)	EUR	Euro government bond yields	1972-09 to 2026-09	649	0
zero_de_4y	German zero-coupon yield, 4 years, month-end (Bundesbank, Svensson method)	EUR	Euro government bond yields	1972-09 to 2026-09	649	0
zero_de_5y	German zero-coupon yield, 5 years, month-end (Bundesbank, Svensson method)	EUR	Euro government bond yields	1972-09 to 2026-09	649	0
tbill1m_return	Return of 1-month US Treasury bills over the month (Kenneth French Data Library)	USD	US short rate (US equity excess return)	1926-07 to 2026-09	1203	0

Notes on the data.

- **Developed markets equity ex US and ex euro area** has no standard index. We build it in the returns section below by taking the euro area out of MSCI World ex USA (both net total return in USD), using the previous month's market caps as weights.
- **Cash rates.** The euro cash rate is 3-month Euribor. We use it for the excess returns (returns in euro minus last month's Euribor divided by 12) and in the currency excess return x_t. The USD cash rate is 3-month USD LIBOR, continued with 3-month Term SOFR plus 0.26161 percentage points after LIBOR ended (see above). We need the USD rate for the interest rate differential in x_t and, in task 4, to convert the US equity view into euro.
- **The equity indices (US, euro area, developed ex US, emerging markets) are net total return:** dividends are reinvested after withholding tax, which is what an institutional investor can replicate.
- **Interior gaps exist only in a predictor** (the euro corporate spread between late 1998 and mid-2000). We handle them where each predictor is used.

Saving data to reuse later.

```
save(panel, core, long_history, series_info, file = "data.RData")
```

```
# Everything below runs from data.RData alone
load("data.RData")
```

Returns

All asset classes are total return indices, so the monthly return is simply the change in the index level: $r_t = P_t/P_{t-1} - 1$. The return of month t is stamped at the end of month t. Because `core` starts at the end of December 1998, the first return is January 1999, which gives 333 monthly returns up to September 2026. At this stage every return is still in the quotation currency of its index (USD or EUR). The step of converting to euro and applying the currency policy comes next.

```r
# monthly simple return of a series of index levels
simple_return <- function(level) level[-1] / level[-length(level)] - 1

# annualized volatility of monthly returns
vol_annual <- function(r) sd(r) * sqrt(12)

# format a decimal as a percentage string, e.g. 0.0712 -> "7.1%"
pct <- function(x, digits = 1) sprintf(paste0("%.", digits, "f%%"), 100 * x)

asset_names <- c(us_equity     = "US equity",
                 euro_equity   = "Euro area equity",
                 dev_equity    = "Developed markets equity ex US and ex euro area",
                 em_equity     = "Emerging markets equity",
                 euro_govt     = "Euro government bonds",
                 us_treasuries = "US Treasuries",
                 euro_corp     = "Euro investment-grade corporate bonds")
asset_currency <- c(us_equity = "USD", euro_equity = "EUR", dev_equity = "USD", em_equity = "USD",
                    euro_govt = "EUR", us_treasuries = "USD", euro_corp = "EUR")

# returns of the indices in their quotation currency
returns_local <- data.frame(date          = core$date[-1],
                            us_equity     = simple_return(core$eq_us_usd),
                            euro_equity   = simple_return(core$eq_emu_eur),
                            em_equity     = simple_return(core$eq_em_usd),
                            euro_govt     = simple_return(core$bd_egov_eur),
                            us_treasuries = simple_return(core$bd_ust_usd),
                            euro_corp     = simple_return(core$bd_ecorp_eur))
```

Developed markets equity ex US and ex euro area

There is no standard index for this asset class (Japan, the UK, Switzerland, Canada, Australia, Denmark, Norway, Sweden, Hong Kong, Singapore, Israel and New Zealand). MSCI World ex USA covers all developed markets except the US, i.e. the euro area plus this rest. It is market-cap weighted, so its return is the weighted average of the two parts:

$$r_t^{\text{World ex US}} = w_{t-1}\, r_t^{\text{EMU}} + (1 - w_{t-1})\, r_t^{\text{rest}}, \qquad w_{t-1} = \frac{\text{market cap EMU}_{t-1}}{\text{market cap World ex USA}_{t-1}}$$

Solving for the part we want gives

$$r_t^{\text{rest}} = \frac{r_t^{\text{World ex US}} - w_{t-1}\, r_t^{\text{EMU}}}{1 - w_{t-1}}.$$

2 important details to note: The weight is taken at the end of the previous month, because a portfolio's return over a month depends on its composition at the start of that month. Furthermore, both returns are in USD (MSCI EMU in USD, not the EUR version used for euro area equity), so the weights refer to the same currency. The result is a USD return, like the other foreign equity classes.

```r
# euro area share of MSCI World ex USA at each month-end
euro_area_weight <- core$mcap_emu / core$mcap_wxus

# weight at the start of each month = previous month-end
weight_start <- euro_area_weight[-length(euro_area_weight)]

# returns of MSCI World ex USA and MSCI EMU, both in USD
world_ex_us_return <- simple_return(core$eq_wxus_usd)
emu_usd_return     <- simple_return(core$eq_emu_usd)
```

```r
# remove the euro area from World ex USA
returns_local$dev_equity <- (world_ex_us_return - weight_start * emu_usd_return) / (1 - weight_start)

# put the columns in SAA order
returns_local <- returns_local[, c("date", names(asset_names))]
```

Returns in euro under the currency policy

The product measures everything in euro. Euro area equity, euro government bonds and euro corporate bonds are already in euro. US equity, developed markets equity ex US and ex euro area, emerging markets equity and US Treasuries are quoted in USD and therefore need to be converted in two steps: convert the value to euro, and then apply the hedge prescribed by the currency policy. Finally, we will express all returns in excess of the euro cash rate.

Exchange rate and currency excess return

Bloomberg quotes EUR/USD as USD per EUR, so the euro price of one dollar is $S_t = 1/\text{EURUSD}_t$. The euro return on holding one dollar during month t is $e_t = S_t/S_{t-1} - 1$, which is positive when the dollar appreciates. The currency excess return is the return of a long position in USD financed in EUR: the dollar's move plus the interest rate differential earned over the month,

$$x_t = e_t + \frac{i_{t-1}^{USD} - i_{t-1}^{EUR}}{12},$$

where i are the 3-month rates at the end of the previous month, in decimals per year. x_t is already an excess return (it is financed in euro), so no cash rate is subtracted from it.

```r
# month t runs from the end of month t-1 to the end of month t
start <- 1:(nrow(core) - 1)
end   <- 2:nrow(core)

# euro price of one dollar (Bloomberg quotes USD per EUR)
eur_per_usd <- 1 / core$eurusd

# e_t: euro return on holding one dollar during month t
usd_return <- eur_per_usd[end] / eur_per_usd[start] - 1

# 3-month rates at the start of each month
euribor_start  <- core$euribor3m[start] / 100
usd_rate_start <- core$usd3m[start] / 100

# x_t: currency excess return of a long USD position financed in EUR
currency_excess <- usd_return + (usd_rate_start - euribor_start) / 12
```

Currency policy

The unhedged euro return of a USD asset is $r_{i,t}^{EUR} = (1 + r_{i,t}^{USD})(1 + e_t) - 1$. Under covered interest parity, hedging a USD asset with a forward contract adds a short position in the currency excess return, so a hedge ratio h_i gives $r_{i,t}^{EUR} - h_i\, x_t$. The current policy hedges bonds in foreign currency fully ($h = 1$ for US Treasuries) and equities quoted in foreign currency by one half ($h = 0.5$ for US equity, developed markets equity ex US and ex euro area, and emerging markets equity). Euro assets need no hedge ($h = 0$).

```r
# SAA weights and hedge ratios of the current currency policy
saa_weight  <- c(us_equity = 0.18, euro_equity = 0.14, dev_equity = 0.07, em_equity = 0.06,
```

```
                euro_govt = 0.25, us_treasuries = 0.10, euro_corp = 0.20)
hedge_ratio <- c(us_equity = 0.5, euro_equity = 0, dev_equity = 0.5, em_equity = 0.5,
                 euro_govt = 0, us_treasuries = 1, euro_corp = 0)
stopifnot(all.equal(sum(saa_weight), 1))

# short USD position of the SAA under the policy (the case states 25.5%)
usd_hedge_saa <- sum(saa_weight * hedge_ratio)

# unhedged euro returns: convert the USD assets with the dollar's move
usd_assets <- names(asset_names)[asset_currency == "USD"]
returns_eur_unhedged <- returns_local
for (a in usd_assets) {
  returns_eur_unhedged[[a]] <- (1 + returns_local[[a]]) * (1 + usd_return) - 1
}

# euro returns under the currency policy: subtract h * x_t
returns_eur_policy <- returns_eur_unhedged
for (a in names(asset_names)) {
  returns_eur_policy[[a]] <- returns_eur_unhedged[[a]] - hedge_ratio[a] * currency_excess
}
```

With the SAA weights, the policy amounts to a short USD position of 25.5% of assets ($0.5 \times (18\% + 7\% + 6\%) + 1 \times 10\%$), which matches the 25.5% stated in the case.

Excess returns

Excess returns are euro returns minus the 3-month Euribor of the previous month-end divided by twelve, the return of holding euro cash over the month. We keep the unhedged version as well, because task 6 regresses the unhedged euro excess returns on x_t.

```
# return of euro cash over month t
cash_return <- euribor_start / 12

# excess returns under the currency policy (task 1 to 5) and unhedged (task 6)
excess_policy   <- returns_eur_policy
excess_unhedged <- returns_eur_unhedged
excess_policy[, names(asset_names)]   <- returns_eur_policy[, names(asset_names)] - cash_return
excess_unhedged[, names(asset_names)] <- returns_eur_unhedged[, names(asset_names)] - cash_return

# the currency excess return is reported alongside the seven asset classes
excess_policy$currency <- currency_excess

stopifnot(!anyNA(excess_policy), !anyNA(excess_unhedged))
```

Summary statistics

All statistics are computed from the 333 monthly excess returns (January 1999 to September 2026) under the current currency policy, and annualized: the mean is the monthly mean times 12, the volatility the monthly standard deviation times $\sqrt{12}$, and the Sharpe ratio their ratio (the returns are already in excess of Euribor, so nothing is subtracted). In addition, the 95% confidence interval of the annualized mean is mean $\pm 1.96 \cdot \sigma / \sqrt{\text{years}}$, and the maximum drawdown is the largest fall of the cumulative excess return from its previous peak.

```
# the seven asset classes plus the currency excess return x_t
stat_series <- c(names(asset_names), "currency")
stat_names  <- c(asset_names, currency = "Currency excess return x (long USD vs. EUR)")
excess      <- excess_policy[, stat_series]
n_years     <- nrow(excess) / 12
```

```r
# cumulative excess return (growth of 1 EUR) and drawdown from the previous peak
cumulative <- as.data.frame(lapply(excess, function(r) cumprod(1 + r)))
drawdown   <- as.data.frame(lapply(cumulative, function(w) w / cummax(w) - 1))

stats <- data.frame(
  mean       = 12 * colMeans(excess),
  volatility = sapply(excess, vol_annual))
stats$sharpe       <- stats$mean / stats$volatility
stats$ci_low       <- stats$mean - 1.96 * stats$volatility / sqrt(n_years)
stats$ci_high      <- stats$mean + 1.96 * stats$volatility / sqrt(n_years)
stats$worst        <- sapply(excess, min)
stats$worst_date   <- excess_policy$date[sapply(excess, which.min)]
stats$max_drawdown <- sapply(drawdown, min)
```

Table 4: Monthly euro excess returns under the current currency policy, annualized, Jan 1999 to Sep 2026

Asset class	Mean p.a.	95% CI of mean	Volatility p.a.	Sharpe ratio	Worst month	Date	Max. drawdown
US equity	6.8%	[1.3%, 12.2%]	14.7%	0.46	-14.0%	Oct 2008	-66.4%
Euro area equity	4.9%	[-1.5%, 11.2%]	17.2%	0.28	-17.8%	Sep 2002	-63.3%
Developed markets equity ex US and ex euro area	4.6%	[-0.4%, 9.6%]	13.5%	0.34	-16.2%	Oct 2008	-54.1%
Emerging markets equity	8.2%	[1.0%, 15.5%]	19.5%	0.42	-25.3%	Oct 2008	-62.1%
Euro government bonds	1.3%	[-0.4%, 3.0%]	4.5%	0.28	-5.2%	Aug 2022	-24.7%
US Treasuries	0.7%	[-1.0%, 2.5%]	4.6%	0.16	-4.6%	Jul 2003	-27.0%
Euro investment-grade corporate bonds	1.6%	[0.1%, 3.1%]	4.1%	0.40	-6.8%	Mar 2020	-16.6%
Currency excess return x (long USD vs. EUR)	1.3%	[-2.1%, 4.8%]	9.3%	0.14	-9.3%	Dec 2008	-46.3%

Figures

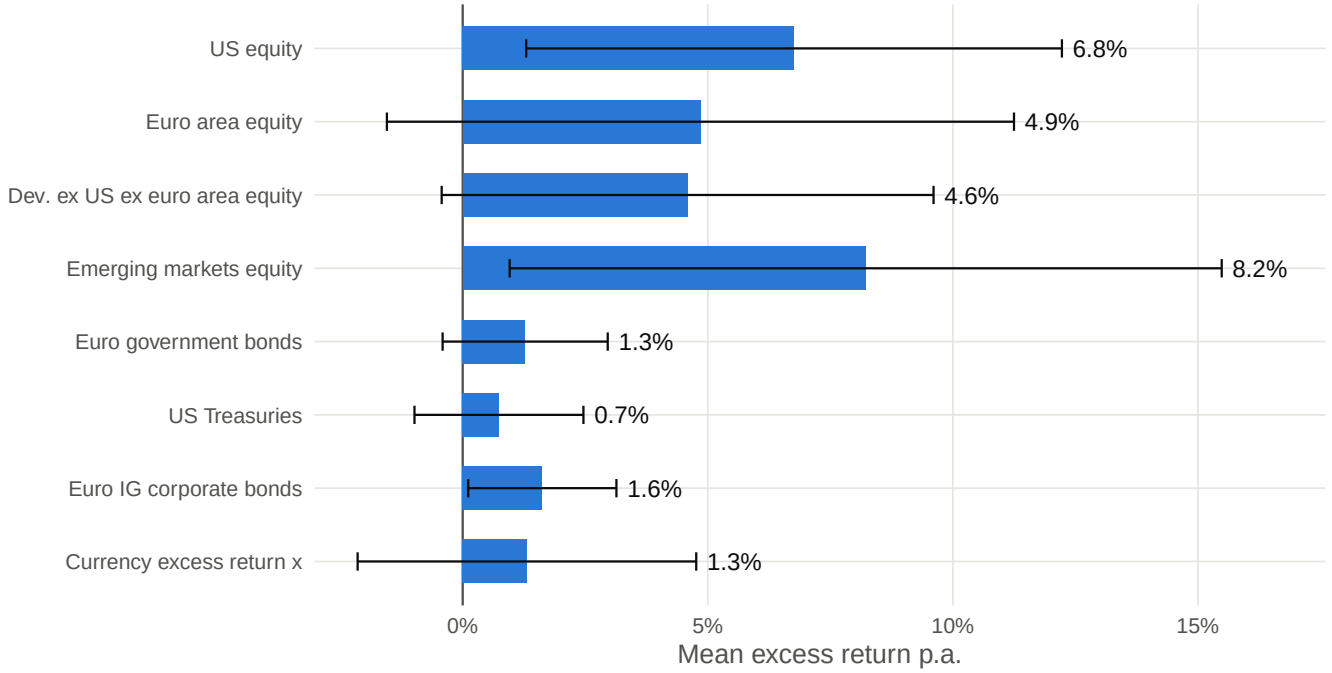


Figure 1: Annualized mean euro excess return under the current currency policy with 95% confidence interval, Jan 1999 to Sep 2026

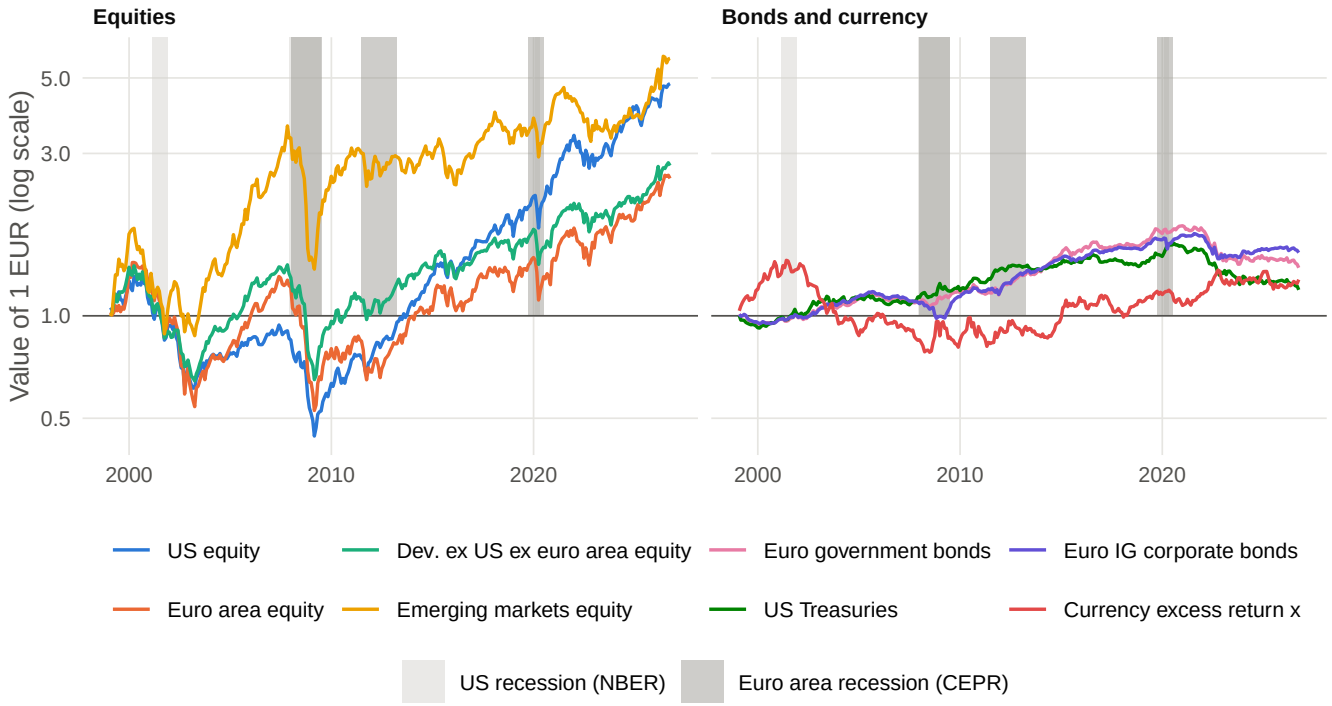


Figure 2: Cumulative euro excess return (growth of 1 EUR invested in excess of Euribor), log scale. Shaded: recessions in the US (NBER) and the euro area (CEPR)

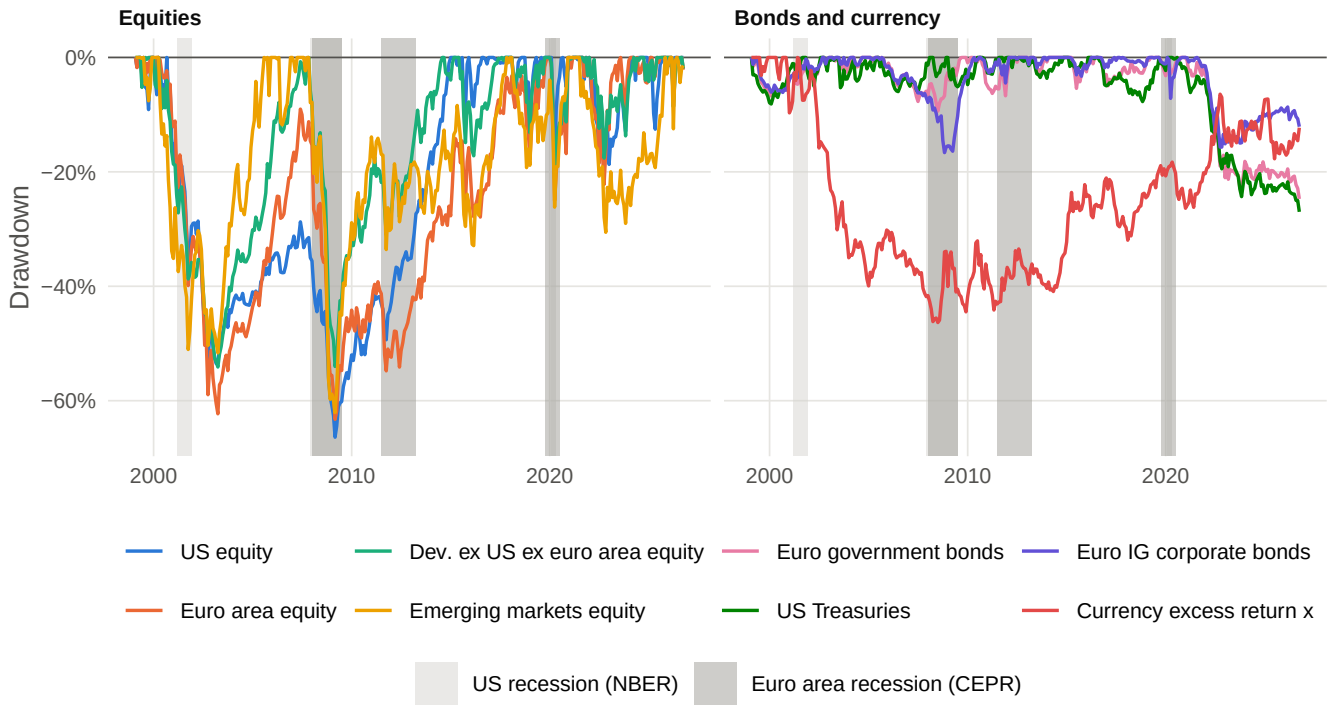


Figure 3: Drawdown of the cumulative euro excess return from its previous peak. Shaded: recessions in the US (NBER) and the euro area (CEPR)

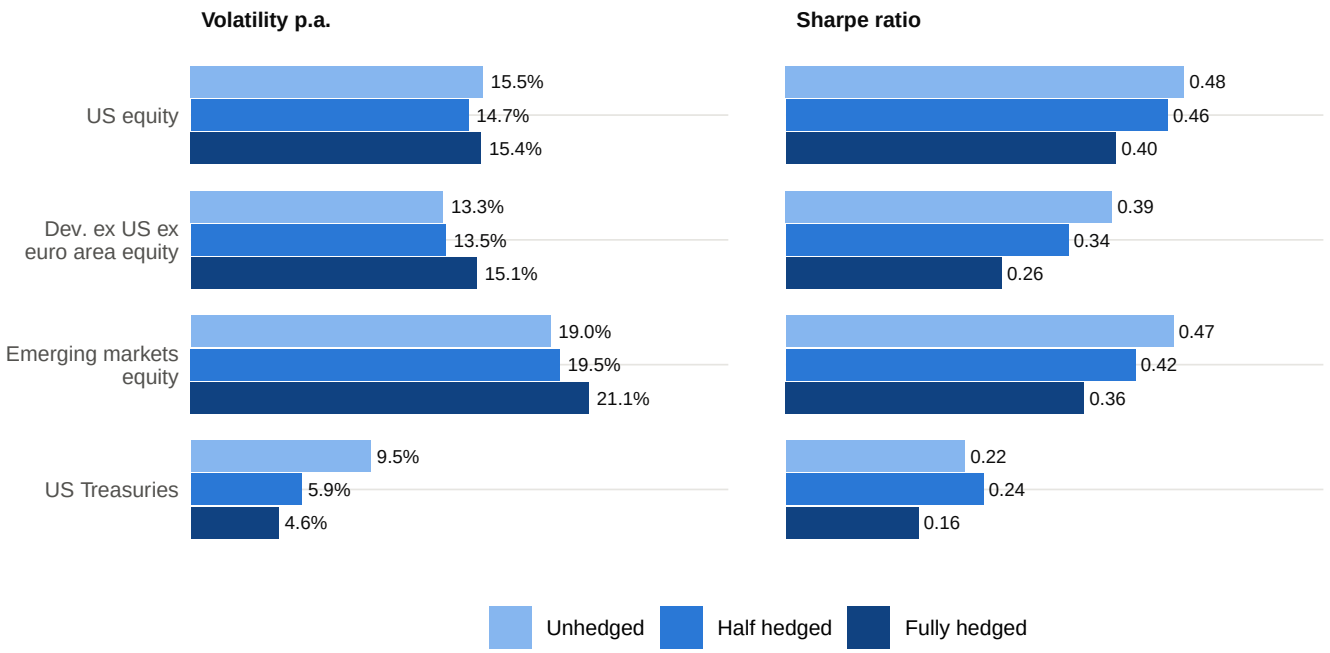


Figure 4: Volatility and Sharpe ratio of the USD assets at different hedge ratios, Jan 1999 to Sep 2026 (current policy: half hedged for equities, fully hedged for US Treasuries)

What stands out

- **Equities earned about 5 times the bond premium, at about 4 times the risk.** Equity excess returns range from 4.6% (developed ex US ex euro area) to 8.2% (emerging markets) per year with volatilities of 13.5% to 19.5%. Bonds earned 0.7% to 1.6% with volatilities around 4 to 5%.
- **US equity has the highest Sharpe ratio (0.46), euro area equity the lowest among equities (0.28).** The euro area suffered from both the 2000 to 2003 bear market and the 2011/12 sovereign debt crisis. Among bonds, euro corporate bonds have the best Sharpe ratio (0.40): they earned the credit spread on top of the government bond return, and their lower volatility is partly due to their shorter duration (4.6 vs. 6.6 years on average), not only to credit.
- **The worst months cluster in crises.** Three of the four equity classes had their worst month in October 2008. Euro corporate bonds had theirs in March 2020 (COVID), and euro government bonds in August 2022 during the 2022 rate shock.
- **Drawdowns highlight the risk.** Equities lost up to 54% to 66% of their cumulative excess value, with deep drawdowns both in 2000 to 2003 and in 2007 to 2009. Government bonds had their largest drawdown not in a crisis but in the rate rises from 2022: the cumulative excess return of euro government bonds is 25% below its peak of December 2020, that of US Treasuries 27% below its peak of July 2020, and neither has recovered by September 2026. In 2022 stocks fell at the same time. Euro corporate bonds had their largest drawdown in the 2008 crisis (17%).
- **The currency hedge matters much more for bonds.** Hedging US Treasuries cuts their volatility from 9.5% unhedged to 4.6%. For equities the half hedge barely changes volatility (US equity 15.5% unhedged vs. 14.7%), and for developed ex US ex euro area and emerging markets equity it even raises it slightly (13.3% vs. 13.5% and 19.0% vs. 19.5%), because the dollar tends to rise when foreign equities fall. The last figure shows this for all three hedge ratios.

Historical means vs. the long run

- **The historical means are estimated very imprecisely.** Even with almost 28 years of data, the 95% confidence intervals for equities are 10 to 15 percentage points wide, and for euro area

equity, developed ex US ex euro area equity, both government bond classes and the currency they include zero. Historical means are therefore a poor forecast of expected returns, which is why the Black-Litterman model starts from equilibrium returns instead.

- **Bonds benefited from falling yields.** The 10-year Bund yield fell from 3.6% in January 1999 to -0.7% in August 2019, adding capital gains that cannot repeat from today's 3.6%. A better anchor is the current yield of each index minus Euribor (2.6%): 1.3% for euro government bonds and 1.6% for euro corporate bonds (before expected default losses).
- **The currency excess return should be about zero in the long run.** Uncovered interest parity does not hold month by month, but there is no reason to expect a lasting premium for holding dollars against euros.
- **The SAA as a whole earned 3.4% per year above Euribor with 7.2% volatility.** That is above the board's long-run assumption of 2.7 percentage points, consistent with the tailwind from falling bond yields, which we do not expect to repeat.

Task 2

Covariance matrix

We estimate the covariance matrix Σ of the seven monthly excess returns (under the current currency policy) over the full common sample, January 1999 to September 2026, and annualize it by multiplying by 12. The full sample is our baseline: it is the longest history available for all seven asset classes and covers several regimes.

```
# monthly excess returns of the seven asset classes as a matrix (columns in SAA order)
excess_matrix <- as.matrix(excess_policy[, names(asset_names)])

# annualized sample covariance over full sample
sigma_full <- cov(excess_matrix) * 12

# ex-ante volatility of a portfolio with weights w
portfolio_vol <- function(w, sigma) sqrt(drop(t(w) %*% sigma %*% w))
saa_vol_full  <- portfolio_vol(saa_weight, sigma_full)
```

The ex-ante volatility of the SAA is $\sqrt{w'_{SAA}\Sigma\, w_{SAA}} = 7.23\%$ per year, below the provider's limit of 9%.

Alternative estimates

As alternatives to the full-sample estimate, we use two methods that put more weight on recent data:

- Estimate the covariance matrix using only the last five years (October 2021 to September 2026) of data.
- Estimate the covariance using an exponentially weighted (EWMA) approach where all months are used, but each month's weight falls by the factor λ per month going back in time. We use a persistence parameter $\lambda = 0.97$, the RiskMetrics (J.P. Morgan) standard for monthly data. It balances responsiveness to new conditions against noise from relying on too few observations. The implied half-life, after which a month's weight has halved, follows from $\lambda^h = 0.5$, i.e. $h = \ln 0.5 / \ln \lambda \approx 23$ months: a return from two years ago counts half as much as last month's.

```
# last five years: sample covariance of the last 60 months
recent_months <- tail(seq_len(nrow(excess_matrix)), 60)
sigma_5y <- cov(excess_matrix[recent_months, ]) * 12

# exponentially weighted covariance
# weight (1 - lambda) * lambda^k for the month k months before the last
ewma_cov <- function(returns, lambda) {
```

```r
  k <- rev(seq_len(nrow(returns)) - 1)            # 0 for the last month, 1 for the month before, ...
  weight <- (1 - lambda) * lambda^k
  weight <- weight / sum(weight)                   # weights sum to one
  demeaned <- sweep(returns, 2, colSums(returns * weight))
  t(demeaned) %*% (demeaned * weight)
}
sigma_ewma <- ewma_cov(excess_matrix, lambda = 0.97) * 12
```

We report each covariance estimate through its two readable parts: the volatilities (square roots of the diagonal of Σ) and the correlations (Σ rescaled by the volatilities, since $\text{cov}_{ij} = \sigma_i \sigma_j \rho_{ij}$).

Table 5: Annualized volatilities of the asset classes and of the SAA under three covariance estimates

Volatility p.a.	Full sample	Last 5 years	EWMA
US equity	14.7%	14.9%	13.7%
Euro area equity	17.2%	14.1%	13.6%
Dev. ex US ex euro area equity	13.5%	12.5%	12.0%
Emerging markets equity	19.5%	16.5%	17.3%
Euro government bonds	4.5%	6.5%	5.4%
US Treasuries	4.6%	5.6%	4.9%
Euro IG corporate bonds	4.1%	5.2%	4.3%
SAA (ex-ante)	**7.23%**	**8.28%**	**7.38%**

Table 6: Selected correlations under three covariance estimates

Correlation	Full sample	Last 5 years	EWMA
Euro area equity / Euro government bonds	0.05	0.59	0.50
US equity / US Treasuries	-0.19	0.42	0.24
Euro area equity / Euro IG corporate bonds	0.36	0.71	0.67
Euro government bonds / Euro IG corporate bonds	0.74	0.92	0.88
US equity / Euro area equity	0.81	0.74	0.72

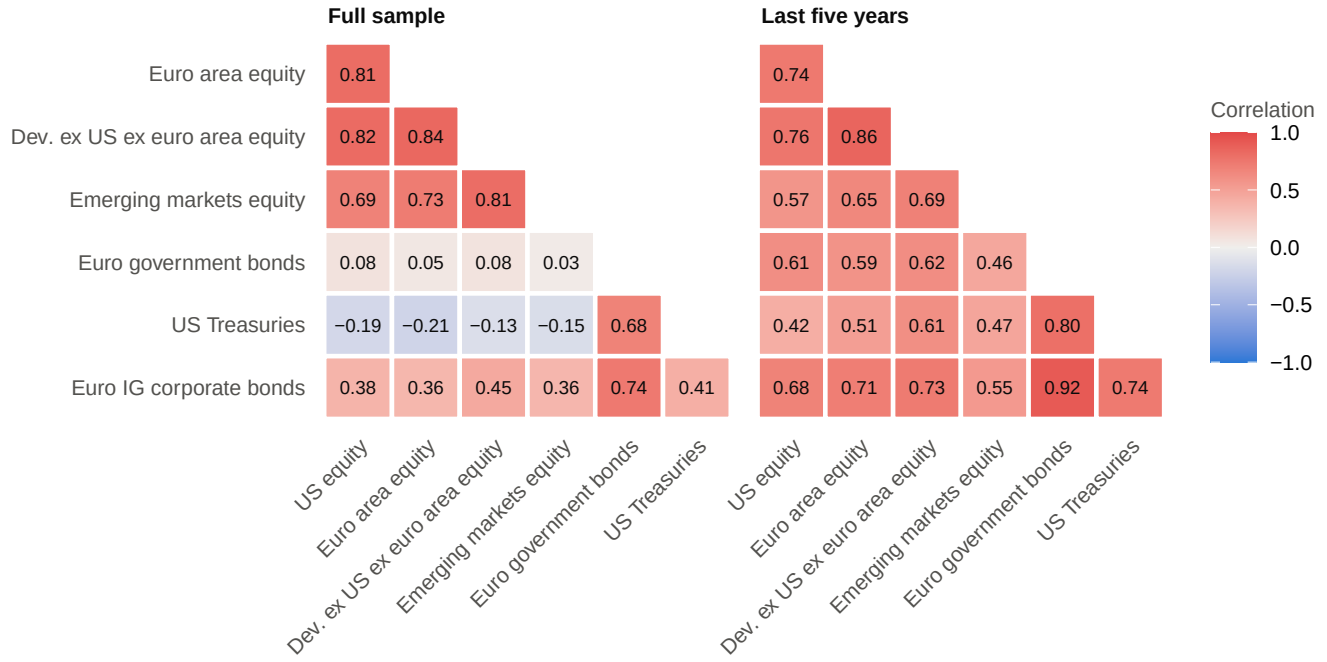


Figure 5: Correlations of the monthly euro excess returns: full sample (Jan 1999 to Sep 2026) and last five years (Oct 2021 to Sep 2026)

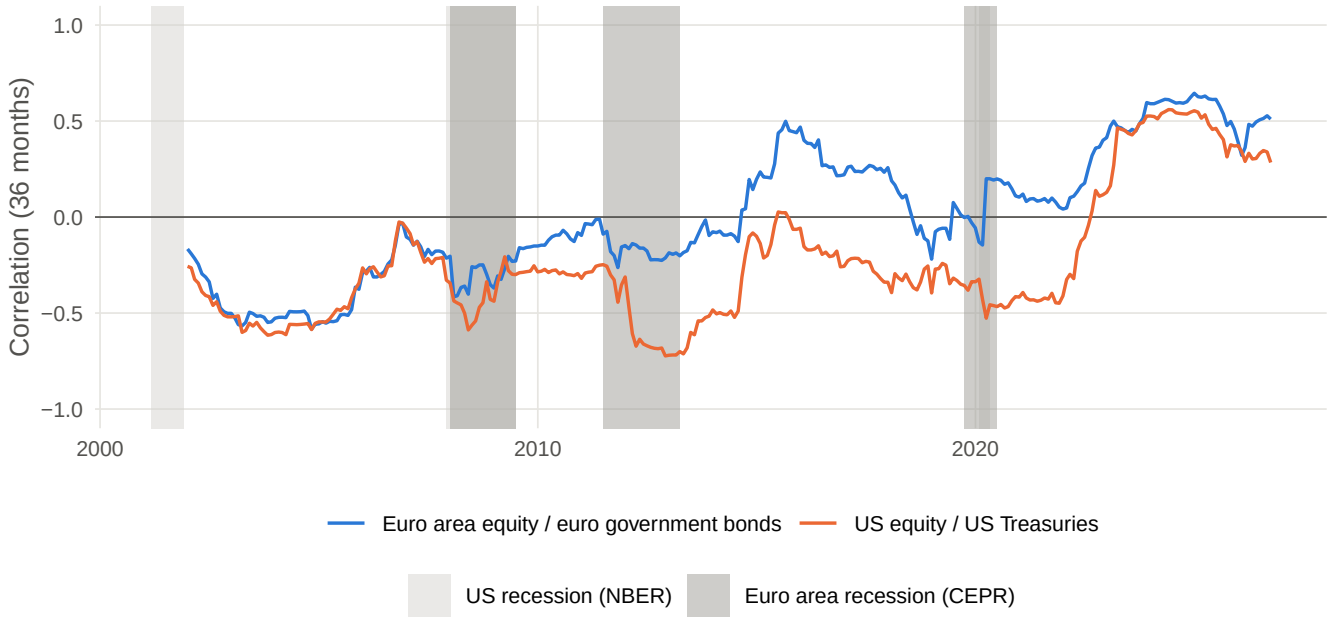


Figure 6: Rolling 36-month correlation between equity and government bond excess returns. Shaded: recessions in the US (NBER) and the euro area (CEPR)

What we would not extrapolate into the coming year

- **The full-sample stock-bond correlation.** Over the full sample, euro area equity and euro government bonds are almost uncorrelated (0.05), and US equity and US Treasuries are negatively correlated (-0.19). That reflects long periods of low inflation, when bonds rallied in equity sell-offs. In the US the correlation was negative until 2021 and has been positive since 2022, when inflation

risk took over. In the euro area it was already mostly positive from 2015, when the ECB's bond purchases lifted both, and it rose further after 2022: the last five years give 0.59 and 0.42. As long as inflation uncertainty stays elevated, the diversification the full sample promises is not there.

- **Low bond volatilities.** Euro government bonds had 4.5% volatility over the full sample, a figure held down by the years of zero interest rates and ECB purchases. Over the last five years it was 6.5%. With yields away from zero, higher bond volatility is thus the better assumption.
- **Crisis-driven equity volatilities in either direction.** The full-sample equity volatilities (17.2% for euro area equity) include 2008 and the euro crisis, while the last five years were calmer (14.1%). Neither should be extrapolated mechanically: crises are rare but occur, so the full-sample equity volatility is a reasonable long-run assumption even if next year turns out calmer.
- **Correlations between government and corporate bonds.** Their correlation rose from 0.74 to 0.92 because rate risk dominated credit risk recently. In a credit event, as in 2008 or March 2020, spreads widen while government bonds rally, and the correlation drops.

With the recent covariance estimates the ex-ante volatility of the SAA rises from 7.23% to 7.38% (EWMA) and 8.28% (last five years). It stays below the 9% limit, but with less room for tactical risk. We nevertheless use the full-sample Σ as the baseline for the equilibrium returns: Π represents the board's long-run view, and the full sample is the only estimate that spans several regimes. Because the coming year may look more like the recent regime, we check the equilibrium returns (task 3) and the volatility limit of the TAA (task 5) with the EWMA Σ, which reflects the positive stock-bond correlation without depending on a single fixed window.

Task 3

Implied equilibrium returns

The board considers the SAA optimal. If an investor with risk aversion A maximizes $w'\mu - \frac{A}{2}w'\Sigma w$, the optimal weights satisfy $\Sigma w = \mu/A$. Turning this around, the expected excess returns under which the SAA is optimal are

$$\Pi = A\,\Sigma\,w_{SAA}.$$

We choose A so that the implied excess return of the SAA equals the board's assumption of 2.7 percentage points. Since $w'_{SAA}\Pi = A\,w'_{SAA}\Sigma\,w_{SAA} = A\,\sigma^2_{SAA}$, this gives $A = 0.027/\sigma^2_{SAA}$. We use the full-sample Σ from task 2.

```
# board's long-run assumption for the SAA excess return
saa_premium <- 0.027

# risk aversion that makes the SAA earn exactly this premium
risk_aversion <- saa_premium / portfolio_vol(saa_weight, sigma_full)^2

# implied equilibrium excess returns
pi_eq <- risk_aversion * drop(sigma_full %*% saa_weight)
stopifnot(all.equal(sum(saa_weight * pi_eq), saa_premium))

# implied Sharpe ratios and the pieces that explain them
vol_full      <- sqrt(diag(sigma_full))
sharpe_eq     <- pi_eq / vol_full
corr_with_saa <- drop(sigma_full %*% saa_weight) / (vol_full * saa_vol_full)
risk_share    <- saa_weight * pi_eq / saa_premium     # share of the SAA variance each asset contributes

# sensitivity: the same calibration with the EWMA covariance from task 2
risk_aversion_ewma <- saa_premium / portfolio_vol(saa_weight, sigma_ewma)^2
```

```
pi_ewma <- risk_aversion_ewma * drop(sigma_ewma %*% saa_weight)
```

The calibration gives $A = 5.17$. The SAA itself then has an implied Sharpe ratio of 0.37 (2.7% divided by its volatility of 7.23%).

Table 7: Implied equilibrium excess returns with the full-sample and the EWMA covariance (risk aversion calibrated to a 2.7 pp SAA premium in both cases). Sharpe ratio, correlation and risk share refer to the full sample

Asset class	SAA weight	Implied (full sample)	Implied (EWMA)	Historical mean	Volatility	Implied Sharpe	Corr. with SAA	Share of SAA risk
US equity	18%	4.94%	4.37%	6.76%	14.7%	0.34	0.90	32.9%
Euro area equity	14%	5.73%	4.39%	4.85%	17.2%	0.33	0.89	29.7%
Developed markets equity ex US and ex euro area	7%	4.52%	3.88%	4.59%	13.5%	0.33	0.90	11.7%
Emerging markets equity	6%	5.81%	4.85%	8.22%	19.5%	0.30	0.80	12.9%
Euro government bonds	25%	0.58%	1.48%	1.27%	4.5%	0.13	0.35	5.4%
US Treasuries	10%	0.06%	0.96%	0.74%	4.6%	0.01	0.04	0.2%
Euro investment-grade corporate bonds	20%	0.96%	1.36%	1.62%	4.1%	0.24	0.63	7.1%

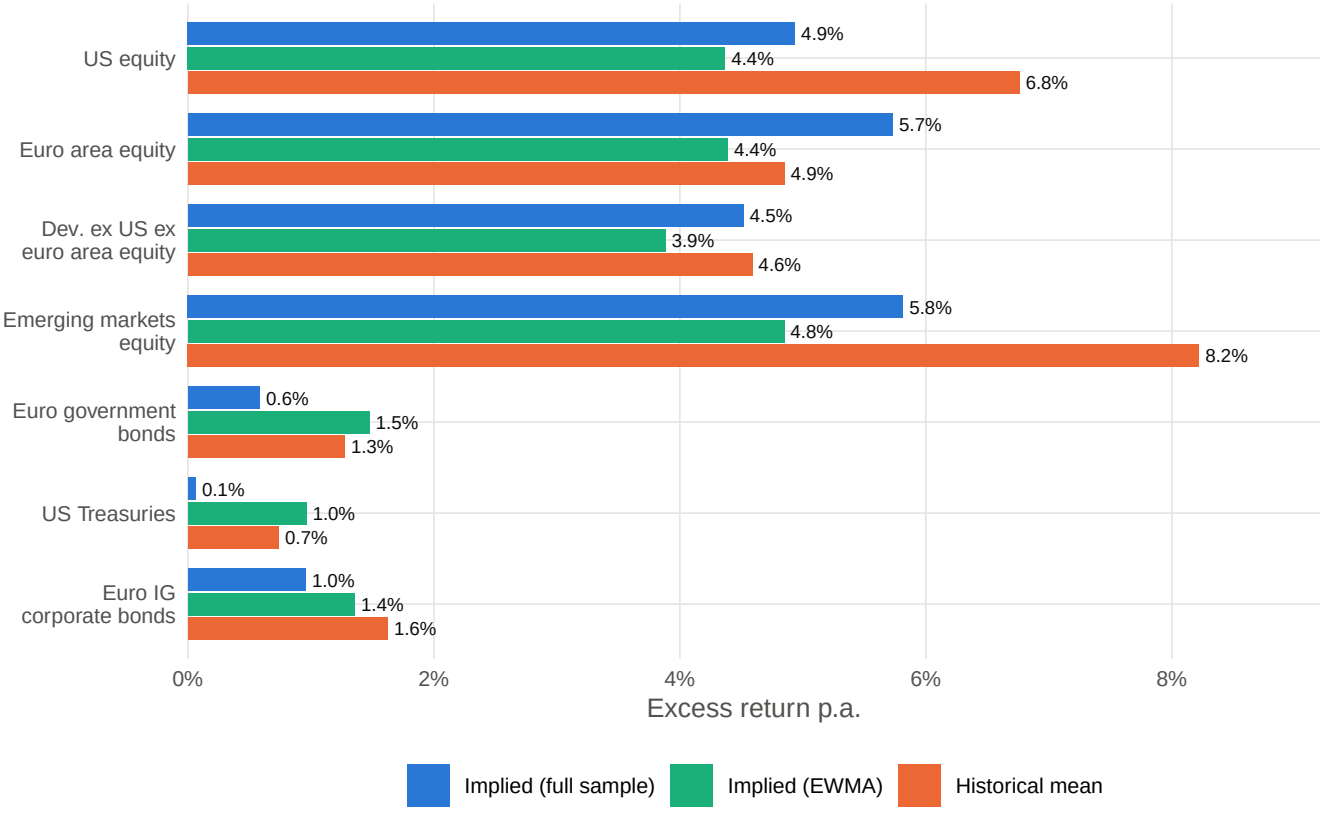


Figure 7: Implied equilibrium excess returns with the full-sample and the EWMA covariance, compared with the historical means (Jan 1999 to Sep 2026)

We can also determine which asset classes have the highest implied Sharpe ratios.

Dividing $\Pi_i = A \operatorname{cov}(r_i, r_{SAA})$ by σ_i shows what drives the implied Sharpe ratio:

$$\frac{\Pi_i}{\sigma_i} = A\,\sigma_{SAA}\,\rho_{i,SAA} = \underbrace{\frac{0.027}{\sigma_{SAA}}}_{\text{Sharpe ratio of the SAA}} \times \rho_{i,SAA}.$$

In equilibrium an asset is only rewarded for the risk it adds to the portfolio the investor holds, so its implied Sharpe ratio is the SAA's Sharpe ratio scaled by its correlation with the SAA.

The equity classes have the highest implied Sharpe ratios (0.30 to 0.34). They are highly correlated with the SAA (0.80 to 0.90) because they dominate its risk: with 45% of the weight they contribute 87% of the SAA variance. US, euro area and developed ex US ex euro area equity have practically the same implied Sharpe ratio; emerging markets are slightly lower because they are less correlated with the rest of the portfolio.

Bonds have low implied Sharpe ratios because they are much less correlated with the SAA, whose risk is mostly equity risk. US Treasuries are the extreme case (correlation 0.04, Sharpe ratio 0.01): over the full sample they rallied when equities fell, so they act as insurance and earn almost no premium. Their low volatility lowers their premia further, but not their Sharpe ratios.

It is however worth noting that the implied bond premia depend heavily on the stock-bond correlation. With the EWMA covariance, which reflects the positive correlation of recent years, the implied excess return of US Treasuries rises from 0.06% to 0.96% and that of euro government bonds from 0.58% to 1.48%. The choice of Σ in task 2 therefore matters for the equilibrium, not only for the volatility limit.

Task 4

Approach

Each view comes from a predictive regression of the excess return over the next twelve months on a predictor observed today,

$$y_{t\to t+12} = a + b\,x_t + \varepsilon_{t+12}, \qquad y_{t\to t+12} = \sum_{j=1}^{12} r^e_{t+j},$$

where r^e are the monthly euro excess returns from task 1. Summing twelve monthly excess returns gives an annual excess return in the same units as Π (twelve times a monthly mean), so the views and the equilibrium returns can be compared directly. The view for the coming year (October 2026 to September 2027) is the fitted value at today's predictor, $v = \hat{a} + \hat{b}\,x_{\text{Sep 2026}}$.

We note the following important points before proceeding with implementation:

- With monthly data and a twelve-month horizon, consecutive observations share eleven months of returns, so the regression errors are autocorrelated. OLS standard errors would be far too small, thus Newey-West standard errors with 12 lags are used.
- We re-run each regression every month with only the data known at that time and forecast the next twelve months. Because predictive regressions often fit well in-sample but fail to beat the historical mean in real time, we compare the squared forecast errors with those of the historical mean: $R^2_{OS} = 1 - \text{MSE}_{\text{model}}/\text{MSE}_{\text{mean}}$. A negative R^2_{OS} means the model did worse than the mean. In that case the in-sample standard error overstates our confidence, so we scale ω_k up by the ratio $\text{MSE}_{\text{model}}/\text{MSE}_{\text{mean}}$.

```r
# excess return over the next 12 months: sum of the monthly excess returns of months t+1 to t+12
next_12m <- function(r) {
  n <- length(r)
  sapply(seq_len(n), function(t) if (t + 12 > n) NA else sum(r[(t + 1):(t + 12)]))
}

# out-of-sample comparison of the model's forecast errors with those of the historical mean
oos_summary <- function(oos) {
  mse_ratio <- sum((oos$actual - oos$model)^2) / sum((oos$actual - oos$hist_mean)^2)
  list(mse_ratio = mse_ratio, r2_oos = 1 - mse_ratio)
}

# predictive regression on consecutive months with Newey-West standard errors
predictive_regression <- function(y, x, dates, oos_start) {
  usable <- which(!is.na(y) & !is.na(x))
  fit <- lm(y[usable] ~ x[usable])
  cov_nw <- sandwich::NeweyWest(fit, lag = 12, prewhite = FALSE)

  # forecast for the next 12 months and its standard error
  now <- max(which(!is.na(x)))
  c_now <- c(1, x[now])
  forecast    <- sum(c_now * coef(fit))
  forecast_se <- sqrt(drop(t(c_now) %*% cov_nw %*% c_now))

  oos <- do.call(rbind, lapply(usable[dates[usable] >= oos_start], function(s) {
    known <- usable[usable <= s - 12]
    b <- coef(lm(y[known] ~ x[known]))
    data.frame(date = dates[s], model = b[1] + b[2] * x[s], hist_mean = mean(y[known]), actual = y[s])
  }))

  c(list(slope = unname(coef(fit)[2]), t_slope = unname(coef(fit)[2] / sqrt(cov_nw[2, 2])),
         r2 = summary(fit)$r.squared, sample = dates[range(usable)], x_now = x[now], x_mean =
         ↪  mean(x[usable]),
         forecast = forecast, forecast_se = forecast_se, oos = oos,
         data = data.frame(date = dates[usable], x = x[usable], y = y[usable])),
    oos_summary(oos))
}

# predictor from the Bloomberg panel, aligned with the months of the excess returns
at_return_dates <- function(series) panel[[series]][match(excess_policy$date, panel$date)]

# first out-of-sample forecast of the Bloomberg-based views: the first estimation window then contains
↪  the 2008/09 crisis
oos_start_bloomberg <- as.Date("2010-01-01")
```

The Bloomberg-based views use the common sample of task 1 (returns from January 1999; the last predictor month with a complete next year of returns is September 2025), with out-of-sample forecasts from January 2010 (the reason is given under the reliability of the views). View 1 uses the full Shiller sample from July 1926. View 2 builds its predictor from German yields since 1972, but its regression on the euro government bond index uses the common sample.

We formulate four views: three absolute views (US equity, euro government bonds, US Treasuries) and one relative view (euro corporate vs. euro government bonds). All views are on euro excess returns under the current currency policy.

View 1: Absolute view on US equity

The first view is an absolute view on US equity based on valuation. When stocks are expensive relative to fundamentals, subsequent returns tend to be low: a valuation ratio can only return to normal through

low prices or high fundamentals, and historically it has mostly been prices. We use the dividend yield D/P (ARM I) and Shiller's CAPE, which divides the price by the 10-year average of real earnings and so smooths out the business cycle, as proxies for valuation. Since expected returns move with yields rather than with price multiples, we use the CAPE yield 1/CAPE.

To run the regressions over a long sample we need monthly US stock returns. For every month since 1871, Shiller reports the S&P Composite index P_t (the average of daily closes over the month rather than the month-end level, which slightly smooths the monthly returns) and the dividends paid on it over the last twelve months D_t (as well as earnings and the CAPE). The S&P Composite stands in for MSCI USA: both are broad market-cap weighted indices of large US companies. A stock's total return consists of the price change plus the dividends received. Since D_t is an annual amount, one month's dividend is roughly $D_t/12$, so the monthly total return is $r_t = (P_t + D_t/12)/P_{t-1} - 1$. Subtracting the return of 1-month US Treasury bills gives the monthly excess return. The T-bill series starts in July 1926, so the sample starts there. Shiller's file currently reports dividends only up to June 2026, so for July to September 2026 we keep the June value. Dividends change very little from month to month, and the trailing dividend yield of MSCI USA in the Bloomberg data (1.06%) is in line with Shiller's (1.07%).

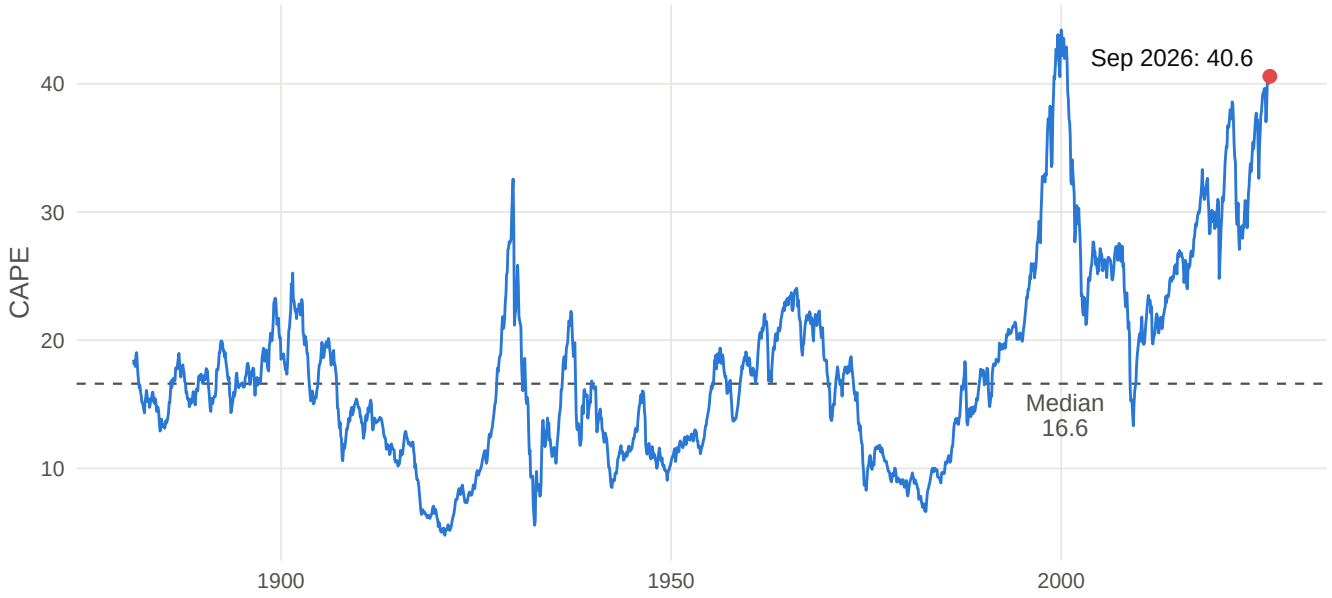


Figure 8: Shiller CAPE of the S&P Composite, 1881 to Sep 2026, with its long-run median.

```
shiller <- long_history[!is.na(long_history$sp_price), ]

last_dividend <- max(which(!is.na(shiller$sp_dividend)))
shiller$sp_dividend[last_dividend:nrow(shiller)] <- shiller$sp_dividend[last_dividend]

n_shiller <- nrow(shiller)
shiller$return <- c(NA, (shiller$sp_price[-1] + shiller$sp_dividend[-1] / 12) /
↪  shiller$sp_price[-n_shiller] - 1)
us_long <- shiller[!is.na(shiller$tbill1m_return), ]
us_long$excess_next_12m <- next_12m(us_long$return - us_long$tbill1m_return / 100)

# predictors: CAPE yield and dividend yield
us_long$cape_yield     <- 1 / us_long$cape
us_long$dividend_yield <- us_long$sp_dividend / us_long$sp_price

reg_cape <- predictive_regression(us_long$excess_next_12m, us_long$cape_yield, us_long$date,
↪  as.Date("1956-01-01"))
```

```
reg_dp   <- predictive_regression(us_long$excess_next_12m, us_long$dividend_yield, us_long$date,
↪  as.Date("1956-01-01"))
```

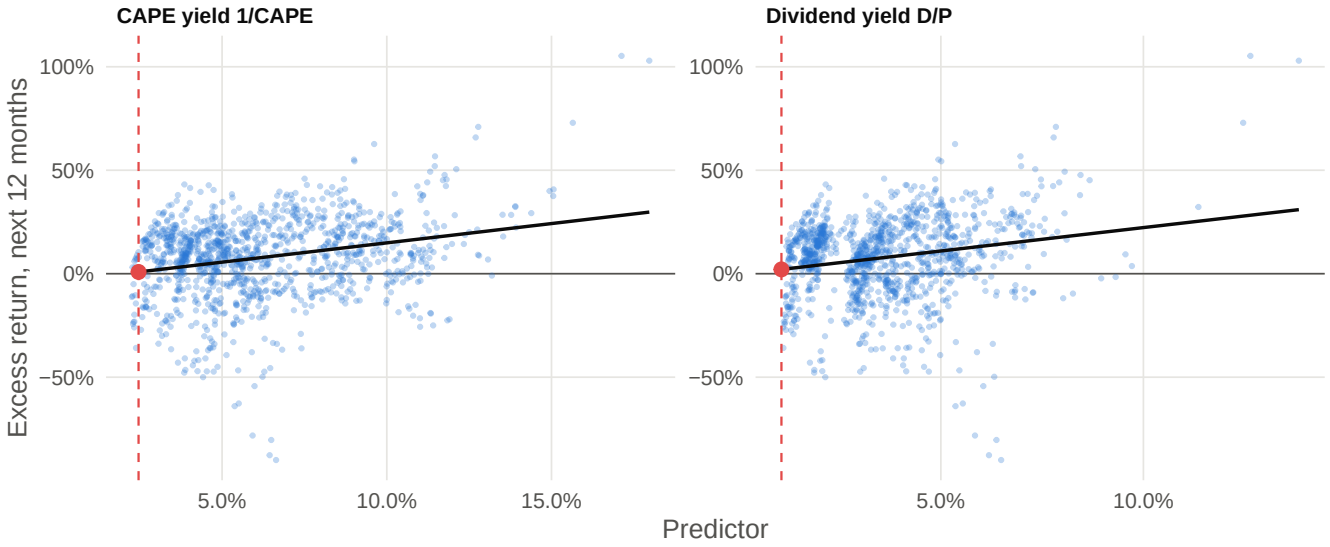


Figure 9: US equity, Jul 1926 to Sep 2025: valuation against the excess return in USD over the next 12 months. Dashed line: predictor at the end of Sep 2026; red dot: the resulting forecast for the next 12 months

The two predictors are highly correlated (0.84), so using them as two separate views would count the same information twice. We combine them into one view by averaging the two forecasts. Averaging forecasts is a simple and robust way to combine predictors (Rapach, Strauss and Zhou 2010). For its standard error we average the two standard errors, which is the upper bound reached if the two estimates were perfectly correlated. Also note that since the 1980s US firms have increasingly paid out cash through buybacks instead of dividends, so recent dividend yields understate the total payout and make the D/P forecast too pessimistic. The CAPE yield is based on earnings and not affected, which is another reason to average the two.

The regression predicts the excess return in USD over 1-month T-bills, whose monthly return we write as rf_t. Under the currency policy, the euro excess return of US equity is exactly

$$r_t^{EUR} - h\,x_t - \frac{i_{t-1}^{EUR}}{12} = \underbrace{\left(r_t^{USD} - rf_t\right)}_{\text{excess return over T-bills}} - \left(\frac{i_{t-1}^{USD}}{12} - rf_t\right) + (1-h)\,x_t + r_t^{USD}\,e_t,$$

so the view in euro has four terms: the regression forecast, minus the gap between the USD cash rate and the T-bill rate (measured today as the 3-month USD rate minus the 3-month T-bill yield), plus the unhedged half of the currency excess return, plus the cross term. For x we assume that the dollar follows a random walk, $E[e] = 0$: the dollar is expected to stay at today's level, so the unhedged half earns the interest rate differential. This is a one-year assumption and differs from the long run in task 1, where we expect the currency excess return to average zero: over short horizons, high-interest currencies have not depreciated as uncovered interest parity predicts (the forward premium puzzle), so the interest rate differential is the better forecast for the coming year.

```
# average of the two forecasts (and of their out-of-sample forecasts)
stopifnot(identical(reg_cape$oos$date, reg_dp$oos$date))
us_equity_oos <- data.frame(date = reg_cape$oos$date, model = (reg_cape$oos$model + reg_dp$oos$model) /
↪  2,
```

```
                          hist_mean = reg_cape$oos$hist_mean, actual = reg_cape$oos$actual)
us_equity_local    <- (reg_cape$forecast + reg_dp$forecast) / 2
us_equity_local_se <- (reg_cape$forecast_se + reg_dp$forecast_se) / 2

# from USD to euro under the currency policy
expected_x <- (tail(core$usd3m, 1) - tail(core$euribor3m, 1)) / 100    # random walk: E[x] = USD rate -
↪  Euribor
cross_term <- 12 * cov(returns_local$us_equity, usd_return)            # covariance of US equity with the
↪  dollar, p.a.
# the regression predicts the excess return over T-bills; the formula needs it over the USD cash rate
cash_gap <- (tail(core$usd3m, 1) - panel$tbill3m[nrow(panel)]) / 100
us_equity_view <- us_equity_local - cash_gap + (1 - hedge_ratio[["us_equity"]]) * expected_x + cross_term
```

View 2: Absolute view on euro government bonds

In ARM I we saw that the shape of the yield curve predicts bond excess returns: Cochrane and Piazzesi (2005) combine the forward rates into a single factor, the CP factor, which forecasts the excess returns of bonds of all maturities. We build it from German zero-coupon yields (Bundesbank, 1 to 5 years, since September 1972), the benchmark curve of the euro area, in two steps:

1. From the yields we compute the five forward rates and, for each month, the excess return of holding a 2- to 5-year German zero-coupon bond for one year instead of the 1-year bond.
2. We regress the average of these excess returns on the five forward rates. The fitted value is the CP factor: the bond excess return that the current shape of the curve predicts.

The euro government bond index is not a German zero-coupon bond (it has a duration of about 6.6 years and also includes Italy, France and Spain). As for the other views, we therefore regress the index's 12-month euro excess return on the CP factor (1999 to 2026), which gives the view and its ω. For simplicity, the out-of-sample check keeps the CP factor estimated on the full German sample, which slightly flatters this view.

```
german <- long_history[!is.na(long_history$zero_de_1y), c("date", paste0("zero_de_", 1:5, "y"))]
n_german <- nrow(german)

# log yields and log bond prices p(n) = -n * y(n), maturities 1 to 5 years in the columns
log_yield <- as.matrix(german[, -1]) / 100
log_price <- -sweep(log_yield, 2, 1:5, "*")

# forward rates: f(1) = y(1), f(n) = p(n-1) - p(n)
forwards <- cbind(log_yield[, 1], log_price[, 1:4] - log_price[, 2:5])

# excess return of buying an n-year bond today and selling it as an (n-1)-year bond in 12 months
bought <- 1:(n_german - 12)
sold   <- 13:n_german
bond_excess <- matrix(NA, n_german, 4)
for (n in 2:5) bond_excess[bought, n - 1] <- log_price[sold, n - 1] - log_price[bought, n] -
↪  log_yield[bought, 1]

# CP factor: fitted value of the average excess return regressed on the five forward rates
cp_fit <- lm(rowMeans(bond_excess) ~ forwards)
german$cp_factor <- drop(cbind(1, forwards) %*% coef(cp_fit))

# view: 12-month excess return of the euro government bond index on the CP factor
govt_bonds <- data.frame(date = excess_policy$date, y = next_12m(excess_policy$euro_govt))
govt_bonds$cp_factor <- german$cp_factor[match(govt_bonds$date, german$date)]
reg_cp <- predictive_regression(govt_bonds$y, govt_bonds$cp_factor, govt_bonds$date, oos_start_bloomberg)
```

The CP factor today is 0.30%. That is below its average of 1.09% since 1972, because the German curve

from 1 to 5 years is almost flat (3.04% to 3.36%). But it is close to its average of 0.34% over the regression sample 1999 to 2025. The view of 1.50% is therefore roughly the historical mean excess return of the index, which includes the gains from falling yields discussed in task 1. We consider it an upper bound.

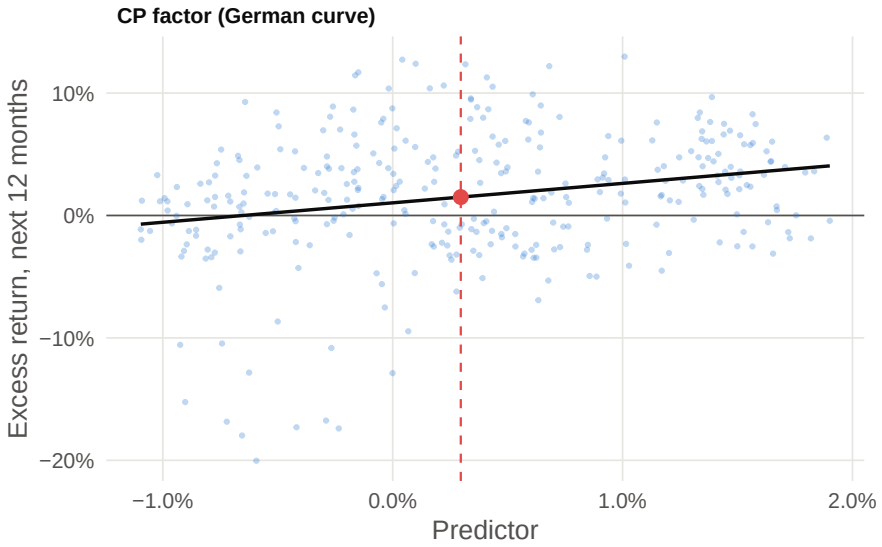


Figure 10: Euro government bonds, Jan 1999 to Sep 2025: CP factor against the euro excess return over the next 12 months. Dashed line: predictor at the end of Sep 2026; red dot: the resulting forecast for the next 12 months

View 3: Absolute view on US Treasuries using the term spread

A steep yield curve predicts high bond excess returns: the long yield contains a term premium that is earned on average (Fama and Bliss 1987, Campbell and Shiller 1991). We use the spread between the 10-year Treasury yield and the 3-month Treasury bill yield. Both are government rates, so unlike a bank rate such as LIBOR, the short leg contains no bank credit risk, which would distort the spread in crises. US Treasuries are fully hedged, so their euro excess return under the policy equals their excess return in USD over the USD cash rate (plus a negligible cross term): for a fully hedged asset, the view in euro and in local currency are the same.

```
treasuries <- data.frame(date = excess_policy$date, y = next_12m(excess_policy$us_treasuries),
                         term_spread = (at_return_dates("y10_us") - at_return_dates("tbill3m")) / 100)
reg_term <- predictive_regression(treasuries$y, treasuries$term_spread, treasuries$date,
↪  oos_start_bloomberg)
```

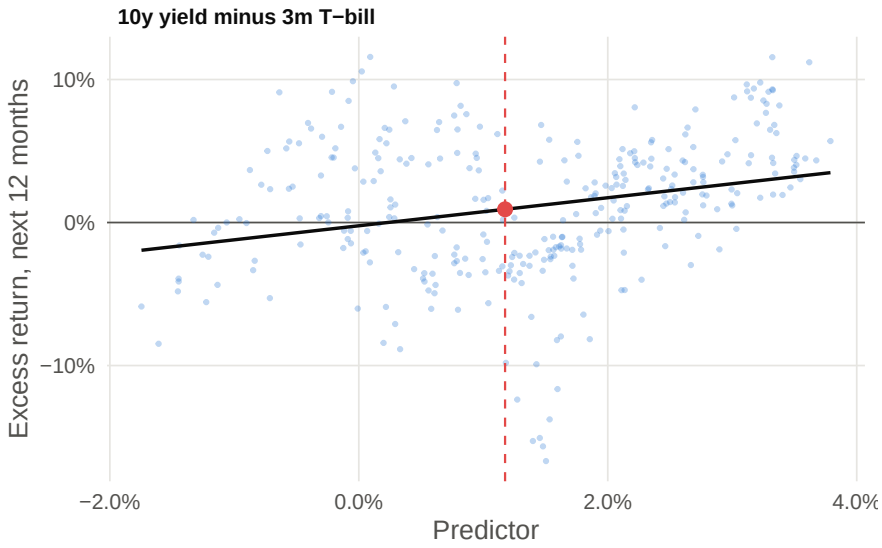


Figure 11: US Treasuries, Jan 1999 to Sep 2025: term spread against the euro excess return over the next 12 months. Dashed line: predictor at the end of Sep 2026; red dot: the resulting forecast for the next 12 months

The term spread today is 1.18%, against an average of 1.36% over 1999 to 2025, so the view of 0.92% is close to the historical mean excess return, which includes the gains from falling yields; we consider it an upper bound. Unlike the CP factor, this regression beat the historical mean out of sample ($R^2_{OS} = 5.3\%$).

View 4: Relative view on euro corporate vs. euro government bonds using the credit spread

The credit spread is the price of credit risk. As the course showed, spreads predict bond returns rather than subsequent defaults: a wide spread means a high expected excess return of corporate over government bonds, a tight spread a low one. The predictor is the option-adjusted spread of the euro corporate index over government bonds.

```
# 12-month excess return of euro corporate over euro government bonds, and the credit spread
credit <- data.frame(date = excess_policy$date,
                     y    = next_12m(excess_policy$euro_corp - excess_policy$euro_govt),
                     oas  = at_return_dates("oas_ecorp") / 100)
reg_oas <- predictive_regression(credit$y, credit$oas, credit$date, oos_start_bloomberg)
```

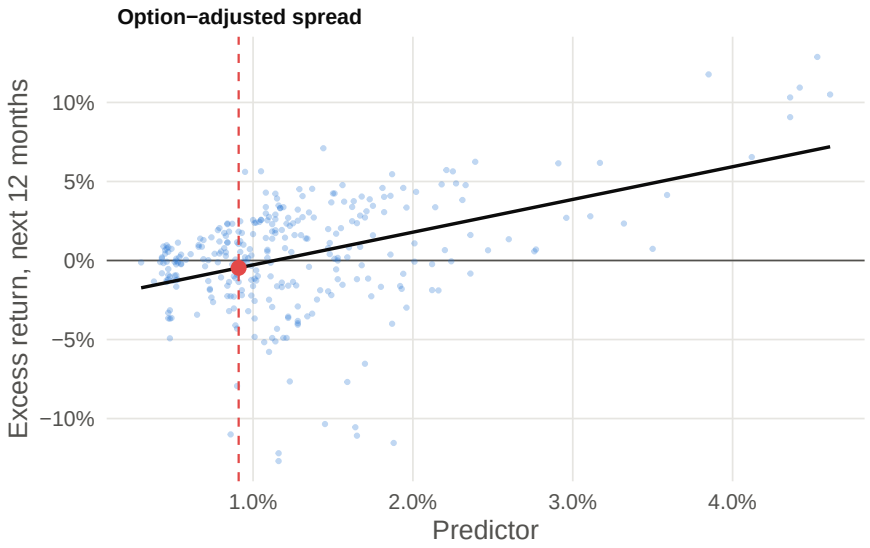


Figure 12: Euro corporate minus government bonds, 1999 to Sep 2025: credit spread against the difference in euro excess returns over the next 12 months. Dashed line: predictor at the end of Sep 2026; red dot: the resulting forecast for the next 12 months

The regression line crosses zero at a spread of about 1.1%: below that, the spread barely compensates for expected default and downgrade losses and for the risk that spreads widen from tight levels. Today's spread of 0.91% (average 1.28%) is below that level, so the view expects corporate bonds to trail government bonds over the coming year (-0.47%). The return difference is not pure credit, however: the corporate index has a shorter duration than the government index (4.3 vs. 6.6 years today), so part of it reflects interest rate moves rather than credit.

Regression results

The table collects all regressions where the following information is presented:

- **Slope** and t (NW) represent the coefficient b and its t-statistic, using Newey-West standard errors.
- R^2 shows the share of the variation in 12-month returns that the regression explains in-sample.
- R^2_{OS} is the out-of-sample R^2 from the "approach" section. Positive here means that the regression forecast the next 12 months better than the historical mean did. Negative means it did worse.
- **Predictor now** and **Predictor avg.** are the predictor at the end of September 2026 and its average over the sample.
- **Forecast**: the expected excess return for October 2026 to September 2027. For US equity it is in USD over 1-month T-bills (converted to euro in view 1). All others are already in euro.

Table 8: Predictive regressions of the 12-month excess return on each predictor

View	Predictor	Sample	Slope	t (NW)	R^2	R^2_{OS}	Predictor now	Predictor avg.	Forecast
1 US equity	CAPE yield 1/CAPE	1926-2025	1.87	2.96	6.7%	-30.5%	2.46%	6.23%	0.85%
1 US equity	Dividend yield D/P	1926-2025	2.26	2.04	4.2%	-5.3%	1.07%	3.63%	2.08%
2 Euro govt bonds	CP factor (German curve)	1999-2025	1.59	1.75	5.9%	-7.1%	0.30%	0.34%	1.50%
3 US Treasuries	10y yield minus 3m T-bill	1999-2025	0.98	2.24	6.8%	5.3%	1.18%	1.36%	0.92%
4 Euro corp minus govt bonds	Corporate OAS	1999-2025	2.07	3.69	19.6%	4.4%	0.91%	1.28%	-0.47%

The views

This section puts the four forecasts into the form $P\mu = v + \varepsilon$ that the Black-Litterman model in task 5 needs. Each row of P states which asset classes a view is about (a single 1 for an absolute view, +1 and -1 for the relative view, so that the row sums to zero), v holds the expected euro excess returns from the regressions, and the diagonal matrix Ω holds the uncertainty ω_k of each view. For comparison, we also compute what the equilibrium of task 3 implies for the same portfolios, $p_k'\Pi$, and how uncertain that prior is, $\tau\, p_k'\Sigma\, p_k$ with $\tau = 0.05$.

```
tau <- 0.05     # scales Sigma into the uncertainty of the equilibrium returns (task 5)

view_ids <- c("us_equity", "euro_govt", "us_treasuries", "corp_minus_govt")

# P: one row per view; an absolute view has a single 1, a relative view goes long one asset and short
↪  another
P <- matrix(0, length(view_ids), length(asset_names), dimnames = list(view_ids, names(asset_names)))
P["us_equity", "us_equity"]         <- 1
P["euro_govt", "euro_govt"]         <- 1
P["us_treasuries", "us_treasuries"] <- 1
P["corp_minus_govt", c("euro_corp", "euro_govt")]  <- c(1, -1)

# v: expected euro excess returns (US equity converted from USD, the others estimated in euro)
v <- c(us_equity = us_equity_view, euro_govt = reg_cp$forecast, us_treasuries = reg_term$forecast,
       corp_minus_govt = reg_oas$forecast)

# omega: variance of the estimated expected return, scaled up if the model lost to the historical mean
↪  out of sample
view_se    <- c(us_equity_local_se, reg_cp$forecast_se, reg_term$forecast_se, reg_oas$forecast_se)
view_oos   <- list(us_equity_oos, reg_cp$oos, reg_term$oos, reg_oas$oos)
mse_ratio  <- sapply(view_oos, function(o) oos_summary(o)$mse_ratio)
omega      <- view_se^2 * pmax(1, mse_ratio)
names(omega) <- view_ids
Omega <- diag(omega, nrow = length(view_ids))

# for comparison: the equilibrium return of each view portfolio and its prior uncertainty tau * p' Sigma
↪  p
pi_view   <- drop(P %*% pi_eq)
prior_var <- tau * diag(P %*% sigma_full %*% t(P))
```

Table 9: Views for Oct 2026 to Sep 2027: expected euro excess returns under the current currency policy, compared with the equilibrium returns of task 3. ω is the variance of the view error in annual units; the last column is the prior uncertainty of the same portfolio for $\tau = 0.05$

View	Type	Predictor (Sep 2026)	View v	Equilibrium $p'\Pi$	$\sqrt{\omega}$	ω	$\sqrt{\tau\, p'\Sigma p}$
US equity	Absolute	CAPE 40.6, D/P 1.07%	1.68%	4.94%	3.00%	0.000899	3.28%
Euro government bonds	Absolute	CP factor 0.30%	1.50%	0.58%	0.93%	0.000086	1.01%
US Treasuries	Absolute	Term spread 1.18%	0.92%	0.06%	0.82%	0.000068	1.04%
Euro corp. minus euro govt bonds	Relative	OAS 0.91%	-0.47%	0.37%	0.54%	0.000029	0.70%

The view matrix in $P\mu = v + \varepsilon$, with the asset classes in SAA order (US equity, euro area equity, developed ex US ex euro area equity, emerging markets equity, euro government bonds, US Treasuries, euro corporate bonds):

Table 10: View matrix P and view vector v

View	US eq.	Euro eq.	Dev. eq.	EM eq.	Euro govt	US Tsy	Euro corp	v
US equity	1	0	0	0	0	0	0	1.68%
Euro government bonds	0	0	0	0	1	0	0	1.50%
US Treasuries	0	0	0	0	0	1	0	0.92%
Euro corp. minus euro govt bonds	0	0	0	0	-1	0	1	-0.47%

What the views say

The views are most negative on US equity: 1.68% against an equilibrium of 4.94%. The CAPE of 40.6 is higher than in 99% of all months since 1881, and both valuation regressions expect a low excess return in USD (1.46% on average). In euro, the unhedged half of the dollar exposure adds 0.86% of interest rate differential, the dollar's covariance with US stocks subtracts 0.41%, and the switch from T-bills to the USD cash rate subtracts 0.24%.

Both bond views lie above equilibrium: 1.50% against 0.58% for euro government bonds and 0.92% against 0.06% for US Treasuries. Their predictors are close to or slightly below their averages, so the regressions expect roughly the usual term premium, while the equilibrium gives bonds little because they hardly add risk to the SAA (task 3).

The relative view is mildly negative. The corporate spread of 0.91% is tight compared with its average of 1.28%, so we expect corporate bonds to trail government bonds (-0.47% against 0.37%).

Reliability of the views

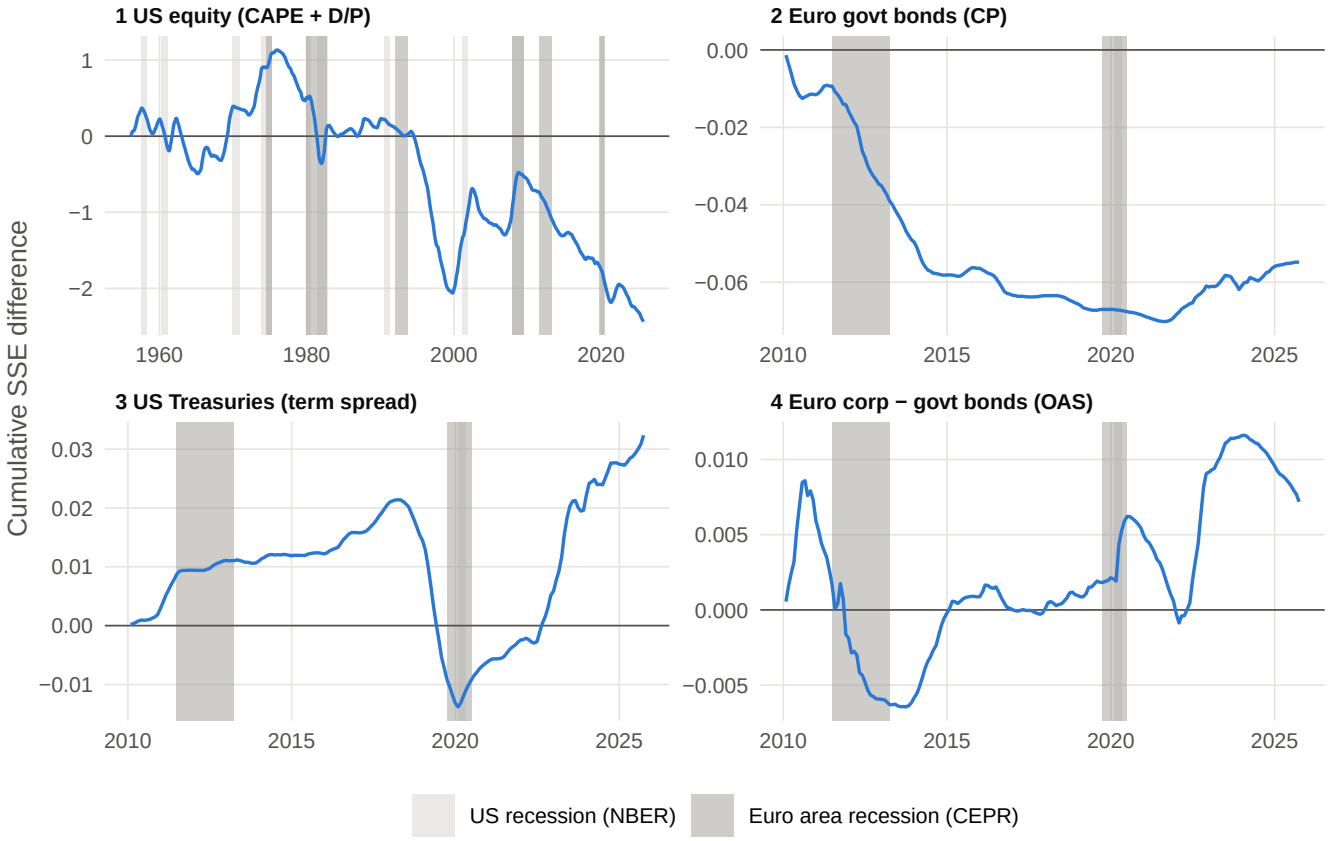


Figure 13: Out-of-sample performance: cumulative squared forecast error of the historical mean minus that of the model. Rising: the model beats the historical mean; falling: it does worse. Shaded: recessions in the US (NBER) and the euro area (CEPR)

Every slope has the expected sign, but only 4 of the 5 are significant at the 5% level (CAPE yield, dividend yield, term spread and credit spread), and the in-sample R^2 of 4% to 20% shows that most of next year's return is noise. That is why a view is a statement about the expected return, not the realized one. Persistent valuation ratios also have upward-biased slopes in small samples (Stambaugh bias), another reason to base the most important view, US equity, on the long Shiller sample.

Out of sample, the term spread (5.3%) and the credit spread (4.4%) beat the historical mean, while the US valuation regressions (combined forecast: -12.2%) and the CP factor (-7.1%) did not. The figure above shows when the latter fell behind: the US valuation regressions from the mid-1990s as valuations kept rising, and the CP factor during the euro crisis of 2010 to 2013. This is the Goyal-Welch critique. We keep these views because each has a clear economic rationale, but scale up their ω by the ratio of the out-of-sample errors.

The out-of-sample forecasts of the three Bloomberg-based views start in January 2010, so that the first estimation window already contains the 2008/09 crisis. Before 2008 the corporate spread only ranged from 0.3% to 1.4%, and a model estimated on that calm decade cannot learn how corporate bonds behave when spreads are wide. With a start in January 2009 the credit spread's R^2_{OS} would be -96.8%, almost entirely because its first forecasts fell into the crisis; the CP factor (-6.2%) and the term spread (5.6%) barely change.

After this adjustment, $\sqrt{\omega}$ is 0.78 to 0.92 times the prior uncertainty $\sqrt{\tau\, p'\Sigma p}$ for every view. Views and

equilibrium therefore get similar weight, and the posterior in task 5 will lie between them rather than jump to the views.

The figure below brings the views and the equilibrium together: for each view portfolio it shows the equilibrium return with its prior uncertainty and the view with its uncertainty ω_k. This is the input to the Black-Litterman update in task 5.

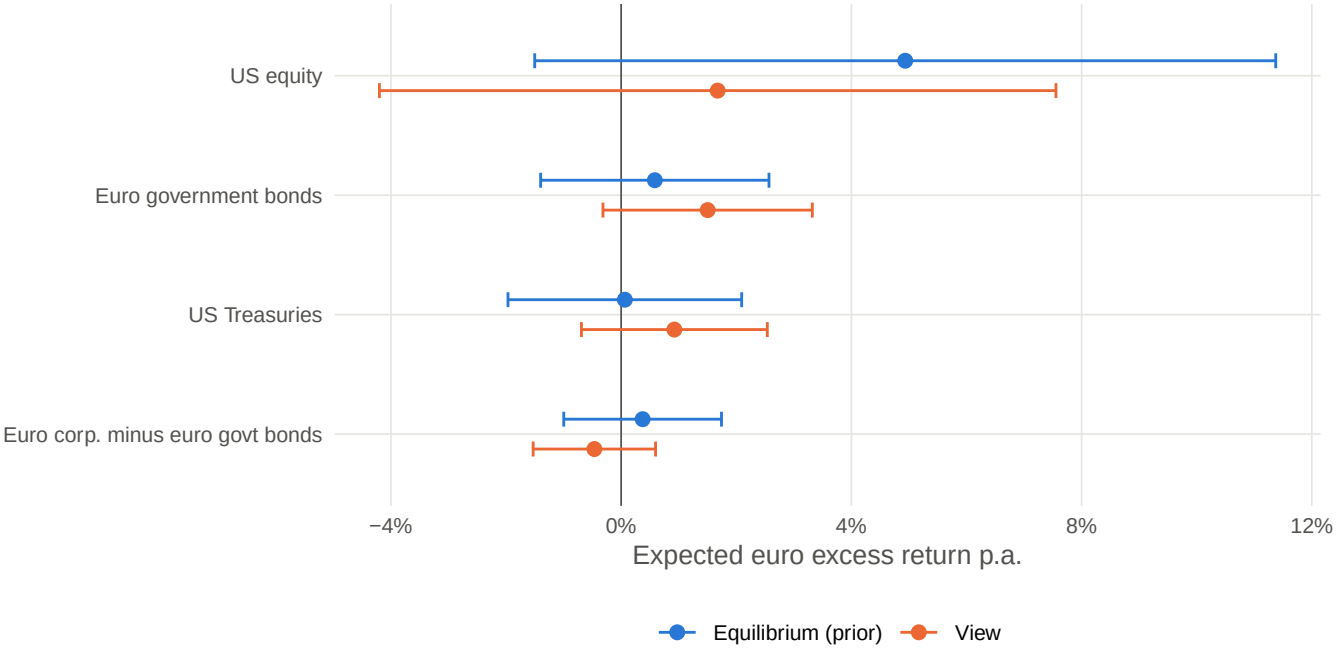


Figure 14: Each view against the equilibrium return of the same portfolio, with 95% intervals: $\pm 1.96\sqrt{\omega}$ for the view and $\pm 1.96\sqrt{\tau\, p'\Sigma p}$ for the equilibrium ($\tau = 0.05$)

Task 5

Black-Litterman posterior

The Black-Litterman model treats the equilibrium returns as a prior, $\mu \sim N(\Pi, \tau\Sigma)$, and the views as a noisy signal, $P\mu = v + \varepsilon$ with $\varepsilon \sim N(0, \Omega)$. Combining the two gives the posterior expected excess returns:

$$\bar{\mu} = \left[(\tau\Sigma)^{-1} + P'\Omega^{-1}P\right]^{-1}\left[(\tau\Sigma)^{-1}\Pi + P'\Omega^{-1}v\right],$$

which is essentially a precision-weighted average of the prior and the views. We use $\tau = 0.05$ and the full-sample Σ for three reasons. It is the same Σ that produced Π in task 3, so prior, posterior and optimization rest on one consistent risk model; a different Σ in this step would shift the expected returns for reasons that have nothing to do with the views. It covers several regimes (2000 to 2003, 2008, the euro crisis, 2022). And the recent estimates would not change whether the volatility limit binds: the TAA stays well below 9% with the EWMA Σ as well (see the discussion below). The same formula can be rewritten as

$$\bar{\mu} = \Pi + \tau\Sigma P'\left(P\tau\Sigma P' + \Omega\right)^{-1}(v - P\Pi),$$

Each view's surprise $v - P\Pi$ (how far it is from equilibrium) is weighted by how confident the view is, and then passed on to every asset class through its covariance with the view portfolio, $\tau\Sigma P'$. This is how asset classes without a view of their own move as well, and it splits $\bar{\mu} - \Pi$ exactly into one contribution per view.

```r
tau_sigma <- tau * sigma_full

# precision-weighted average of the equilibrium (prior) and the views
posterior_precision <- solve(tau_sigma) + t(P) %*% solve(Omega) %*% P
mu_bar <- drop(solve(posterior_precision, solve(tau_sigma) %*% pi_eq + t(P) %*% solve(Omega) %*% v))

# surprise v - P Pi, weighted by its confidence
view_weight  <- drop(solve(P %*% tau_sigma %*% t(P) + Omega, v - drop(P %*% pi_eq)))
contribution <- tau_sigma %*% t(P) %*% diag(view_weight)      # rows: asset classes, columns: views
colnames(contribution) <- view_ids
stopifnot(all.equal(pi_eq + rowSums(contribution), mu_bar))
```

Table 11: Equilibrium and posterior expected euro excess returns ($\tau = 0.05$, full-sample Σ)

Asset class	Equilibrium Π	Posterior $\bar{\mu}$	Change	Sharpe (Π)	Sharpe ($\bar{\mu}$)
US equity	4.94%	2.87%	-2.07%	0.34	0.20
Euro area equity	5.73%	3.51%	-2.22%	0.33	0.20
Developed markets equity ex US and ex euro area	4.52%	2.73%	-1.79%	0.33	0.20
Emerging markets equity	5.81%	3.50%	-2.32%	0.30	0.18
Euro government bonds	0.58%	1.24%	0.65%	0.13	0.27
US Treasuries	0.06%	0.80%	0.74%	0.01	0.17
Euro investment-grade corporate bonds	0.96%	0.98%	0.03%	0.24	0.24

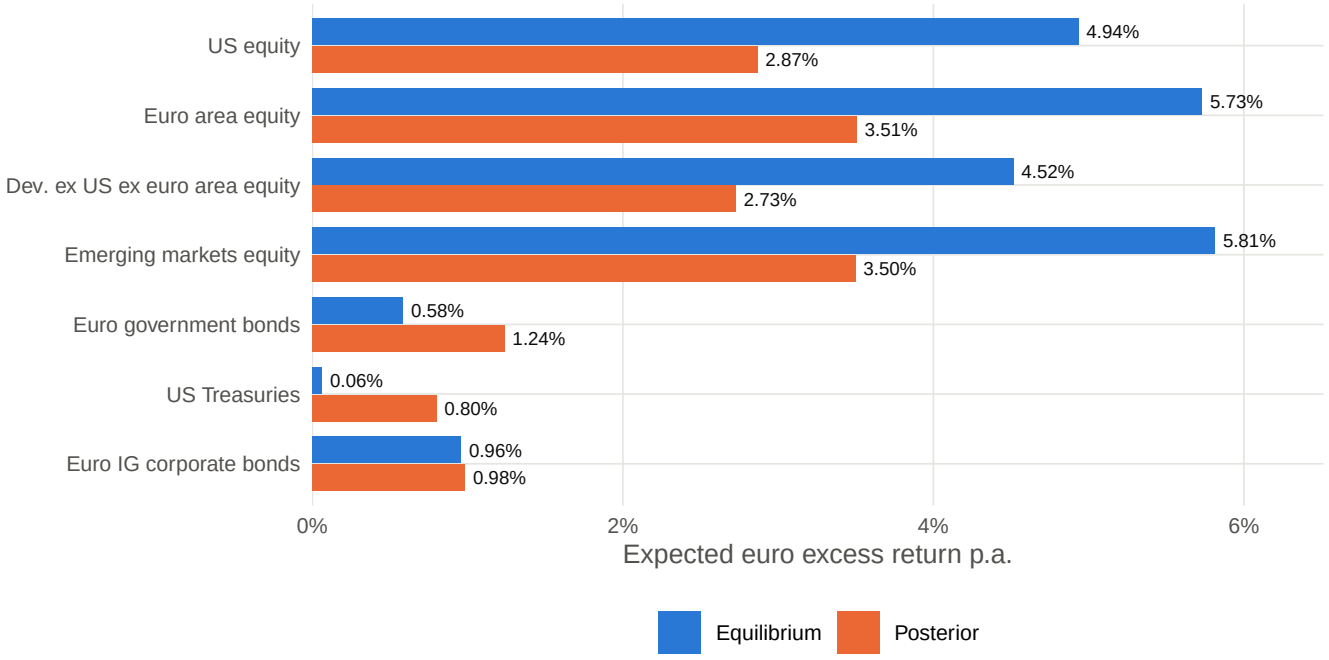


Figure 15: Equilibrium returns Π and posterior expected returns $\bar{\mu}$ after adding the four views

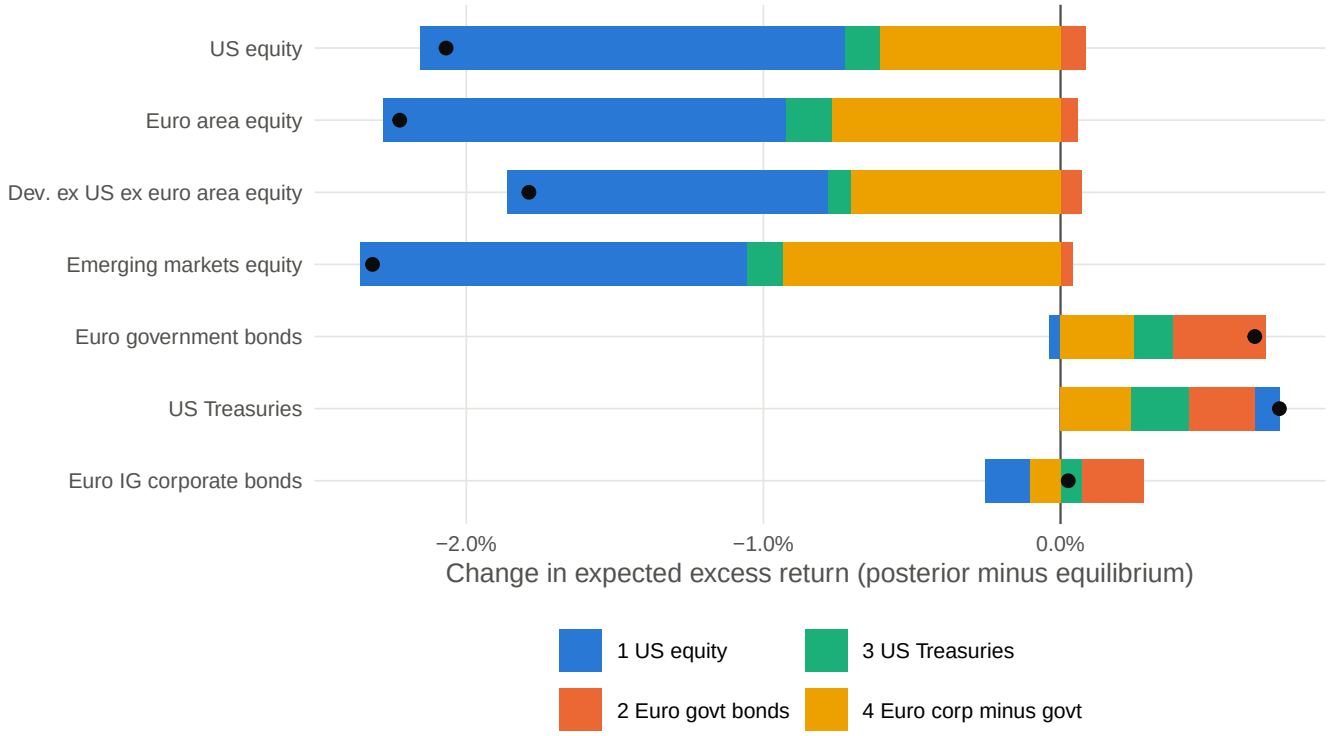


Figure 16: Which views move which asset classes: contribution of each view to the change from Π to $\bar{\mu}$ (bars add up to the black dot)

Which views move which assets

All four equity classes lose 1.8% to 2.3% of expected excess return, although only US equity has a view. The US equity view is the main driver: equity classes are highly correlated, so a lower expected return for US equity also lowers it for euro area, developed and emerging markets equity. Euro area equity, which has no view of its own, falls by 2.22%, more than US equity itself (2.07%).

The credit view also lowers all equities, by 0.6% to 0.9%: corporate minus government bonds is a bet on credit risk, which moves with equities, so a lower expected credit premium implies lower equity returns too. It lowers euro area, developed ex US ex euro area and emerging markets equity more than US equity, which is one reason why euro area and emerging markets equity fall more than US equity. Because government bonds are the short leg of the same view, it also raises the expected return of euro government bonds (0.25%) and, through their correlation, of US Treasuries (0.24%).

The two bond views raise the expected returns of both government bond classes and, through their correlation, of euro corporate bonds. The US Treasury view also lowers equities slightly, because over the full sample Treasuries were negatively correlated with equities.

The posterior lies between equilibrium and views, as the similar uncertainties in task 4 suggested. For US equity, for example, it is 2.87%, between the equilibrium of 4.94% and the view of 1.68%. As a result, euro government and euro corporate bonds now have higher Sharpe ratios than any equity class, while in equilibrium equities had the highest.

Tactical asset allocation

The TAA maximizes $w'\bar{\mu} - \frac{A}{2}w'\Sigma w$ subject to the constraints set by the case (cash at most 5% is the same as $\sum_i w_i \geq 0.95$):

- each asset class within its tactical band around the SAA weight (given in Table 1 of the case)
- total equity within $45\% \pm 8$ percentage points
- $0.95 \leq \sum_i w_i \leq 1$, the rest held in cash (which has a zero excess return)
- an ex-ante volatility of at most 9%

We start from the risk aversion of task 3, $A = 5.17$, and would raise it step by step until the volatility limit is met. Since the problem is quadratic, we use solve.QP to solve it. It minimizes $\frac{1}{2}w'Dw - d'w$ subject to linear constraints $C'w \geq b$, so we set $D = A\Sigma$, $d = \bar{\mu}$ and write every constraint as "greater or equal".

```
tactical_band <- c(us_equity = 0.05, euro_equity = 0.05, dev_equity = 0.03, em_equity = 0.03,
                   euro_govt = 0.05, us_treasuries = 0.04, euro_corp = 0.05)
is_equity <- as.numeric(names(saa_weight) %in% equity)

# optimal TAA weights for a given risk aversion
taa_weights <- function(risk_aversion_taa) {
  n_assets <- length(saa_weight)

  # each column of constraint_matrix is one constraint: constraint_matrix' w >= lower_limit
  constraint_matrix <- cbind(diag(n_assets), -diag(n_assets),
                             is_equity, -is_equity,
                             1, -1)
  lower_limit <- c(saa_weight - tactical_band, -(saa_weight + tactical_band),
                   0.45 - 0.08, -(0.45 + 0.08),
                   0.95, -1)
  solution <- quadprog::solve.QP(Dmat = risk_aversion_taa * sigma_full, dvec = mu_bar,
                                 Amat = constraint_matrix, bvec = lower_limit)$solution
  setNames(solution, names(saa_weight))
}

# start at the calibrated risk aversion and raise it by 5% steps until the volatility limit holds
risk_aversion_taa <- risk_aversion
w_taa <- taa_weights(risk_aversion_taa)
while (portfolio_vol(w_taa, sigma_full) > 0.09) {
  risk_aversion_taa <- risk_aversion_taa * 1.05
  w_taa <- taa_weights(risk_aversion_taa)
}

# portfolio statistics of the SAA and the TAA under the posterior returns
portfolio_stats <- function(w) {
  c(expected = sum(w * mu_bar), vol = portfolio_vol(w, sigma_full), vol_ewma = portfolio_vol(w,
  ↪  sigma_ewma),
    equity = sum(w[equity]), cash = max(0, 1 - sum(w)))      # max(): rounding noise of the optimizer
}
stats_saa <- portfolio_stats(saa_weight)
stats_taa <- portfolio_stats(w_taa)
```

Table 12: Tactical asset allocation for Oct 2026 to Sep 2027

Asset class	SAA	Tactical band	TAA	Deviation	At band	Posterior $\bar{\mu}$
US equity	18%	13% to 23%	13.1%	-4.9 pp		2.87%
Euro area equity	14%	9% to 19%	10.9%	-3.1 pp		3.51%
Developed markets equity ex US and ex euro area	7%	4% to 10%	10.0%	+3.0 pp	upper	2.73%
Emerging markets equity	6%	3% to 9%	3.0%	-3.0 pp	lower	3.50%
Euro government bonds	25%	20% to 30%	30.0%	+5.0 pp	upper	1.24%
US Treasuries	10%	6% to 14%	14.0%	+4.0 pp	upper	0.80%
Euro investment-grade corporate bonds	20%	15% to 25%	19.0%	-1.0 pp		0.98%
Total equity	45%	37% to 53%	37.0%	-8.0 pp	lower	
Cash	0%	0% to 5%	0.0%	+0.0 pp		0.00%

Table 13: Expected excess return and risk of the SAA and the TAA

	SAA	TAA
Expected excess return (posterior)	1.99%	1.81%
Volatility (full-sample Σ)	7.23%	6.11%
Volatility (EWMA Σ)	7.38%	6.65%
Sharpe ratio	0.28	0.30
Utility $w'\bar{\mu} - \frac{A}{2}w'\Sigma w$	0.64%	0.84%

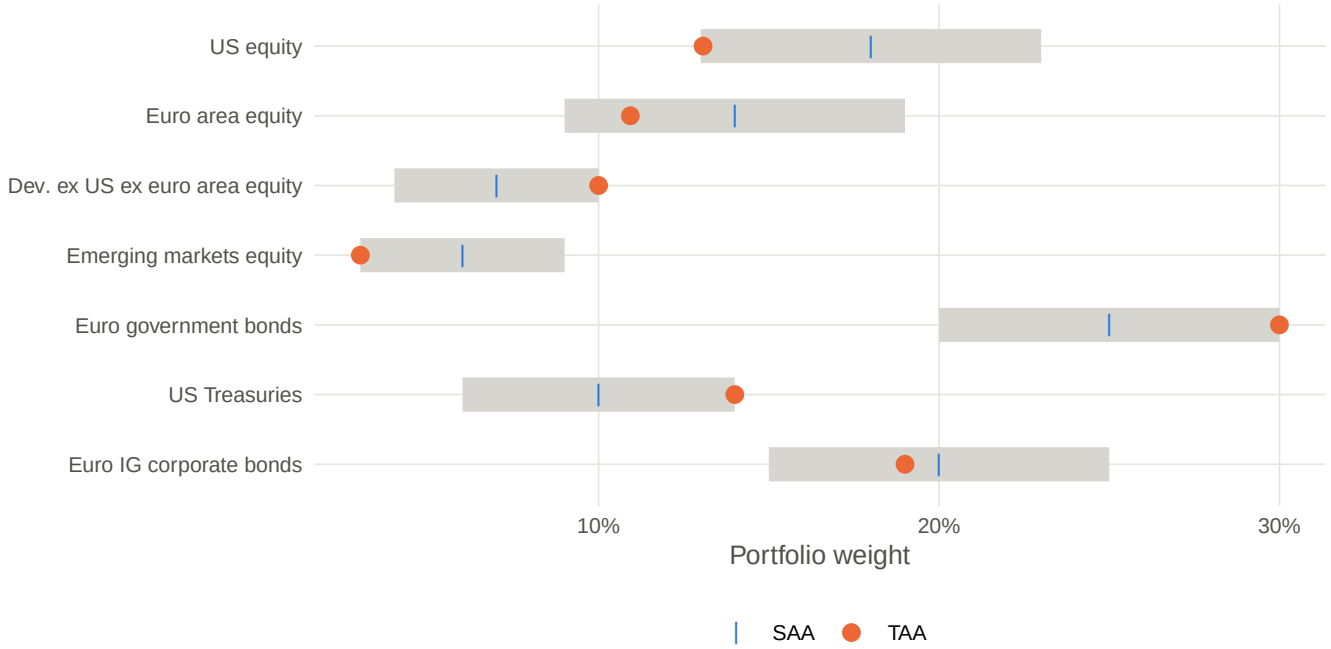


Figure 17: TAA weights within the tactical bands (grey) compared with the SAA weights

Discussion

The volatility limit does not bind. At the calibrated risk aversion the TAA has an ex-ante volatility of 6.11%, well below 9%, and even with the EWMA covariance from task 2 it stays at 6.65%. There is therefore no need to raise A (it stays at 5.17).

The TAA de-risks. With the posterior returns, every equity class earns less than its marginal contribution to risk costs at the board's risk aversion, so the optimizer cuts total equity to the lower limit of 37%. The cut comes mostly from US equity, which ends just above its lower band (-4.9 pp) because it carries the most negative view, and partly from euro area (-3.1 pp) and emerging markets equity (-3.0 pp). Developed ex US ex euro area equity goes to its upper band: it has the lowest volatility of the equity classes, so within the minimum equity share it is the cheapest way to hold equity risk. The money goes into euro government bonds and US Treasuries, both at their upper bands, while euro corporate bonds take the rest (-1.0 pp) because the credit view lowered their advantage over government bonds. The TAA is fully invested (cash 0%).

The TAA expects 1.81% excess return against 1.99% for the SAA (both under the posterior), but at 6.11% instead of 7.23% volatility. Giving up 0.19 pp of expected return for 1.1 pp less volatility raises the Sharpe ratio and the board's utility: once the views have lowered the expected equity premium, the equity risk is no longer worth it.

Most weights sit at a band, so the TAA is driven by the direction of the views more than by their exact size.

Task 6

Hedge regressions

A euro investor who holds a US dollar asset without a hedge implicitly holds a long position in the dollar (with excess return x_t). In order to hedge this exposure, one needs to sell dollars forward, i.e. adding a short position in x_t. This dollar exposure risk depends on how the asset's euro return moves with x_t. We therefore regress the unhedged euro excess return of each asset class on the currency excess return in order to measure how much of each asset's euro return moves with the dollar, which tells us how much of its dollar exposure should be hedged to minimize risk. The regression formula is given as

$$r_{i,t}^{unh} = \alpha_i + \beta_i \, x_t + \varepsilon_{i,t}.$$

Hedging a fraction h of the exposure changes the variance of the asset's return to

$$\mathrm{Var}(r_i^{unh} - h\,x) = \mathrm{Var}(r_i^{unh}) + \mathrm{Var}(x)\,(h^2 - 2h\beta_i),$$

which is smallest at $h = \beta_i$. The slope β_i is therefore the variance-minimizing hedge ratio: holding $r_{i,t}^{unh} - \beta_i x_t$ removes exactly the part of the asset's risk that moves with the dollar. The formula also shows that a hedge adds risk once $h > 2\beta_i$. The regressions use the monthly returns of January 1999 to September 2026 and since the months do not overlap, OLS standard errors are appropriate.

```
# slope of each unhedged euro excess return on the currency excess return x_t
hedge_beta <- t(sapply(names(asset_names), function(a) {
  fit <- lm(excess_unhedged[[a]] ~ currency_excess)
  c(beta = unname(coef(fit)[2]), se = unname(sqrt(vcov(fit)[2, 2])), r2 = summary(fit)$r.squared)
}))
```

Table 14: Regressions of the unhedged euro excess returns on the currency excess return x_t, Jan 1999 to Sep 2026

Asset class	SAA weight	Policy hedge ratio	β_i	t-stat.	R^2	USD position $-w_i\beta_i$
US equity	18%	0.50	0.52	5.96	9.7%	-9.3%
Euro area equity	14%	0.00	-0.25	-2.45	1.8%	3.5%
Developed markets equity ex US and ex euro area	7%	0.50	0.21	2.63	2.0%	-1.4%
Emerging markets equity	6%	0.50	0.03	0.23	0.0%	-0.2%
Euro government bonds	25%	0.00	-0.03	-0.94	0.3%	0.6%
US Treasuries	10%	1.00	0.90	33.49	77.2%	-9.0%
Euro investment-grade corporate bonds	20%	0.00	-0.08	-3.22	3.0%	1.5%

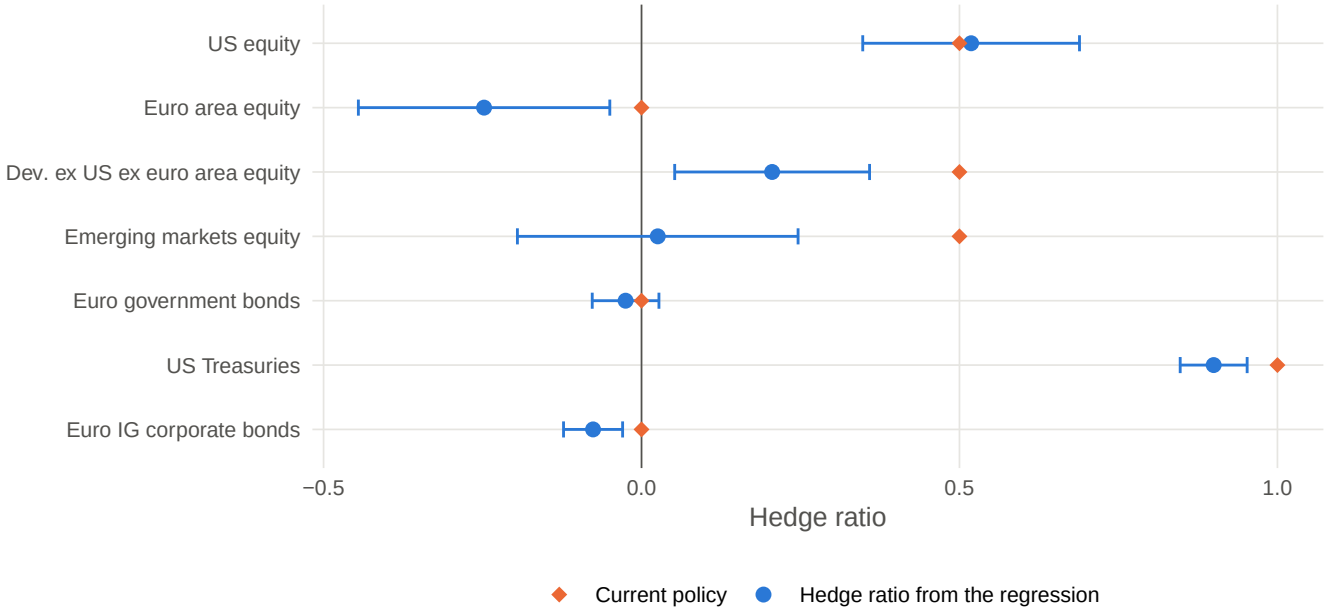


Figure 18: Variance-minimizing hedge ratios β_i with 95% confidence intervals, compared with the hedge ratios of the current currency policy

The regressions confirm that the currency risk depends on co-movement, not on the quotation currency. US Treasuries have a β of 0.90, close to the full hedge of the policy: their euro return is almost entirely bond return plus dollar. For US equity the regression gives 0.52, very close to the policy's half hedge. It is well below 1 because US stocks tend to fall when the dollar rises (flight to safety), which partly offsets the currency exposure.

For developed ex US ex euro area equity (0.21) and emerging markets equity (0.03) it is much lower than 0.5. These indices are quoted in USD, but the underlying stocks are priced in yen, pounds, zloty etc. When the dollar rises their USD returns fall through the exchange rate of their local currencies against the dollar which offsets the dollar's gain on the index from exchanging the USD index into EUR. Their actual exposure is these currencies against the euro and these currencies only move partly with the dollar, thus hedging them with dollar forwards adds risk instead of removing it. The policy's hedge of 0.5 exceeds $2\beta_i$ for both classes, which is why their volatility rose under the half hedge in task 1.

Euro assets are quoted in euro but still have small, mostly negative betas, euro area equity in particular (-0.25): it tends to fall when the dollar rises, so a long dollar position would reduce its risk here.

Variance-minimizing USD position of the SAA

Starting from the all-unhedged SAA, a USD position h (negative for a short) gives the portfolio return $r^{unh}_{SAA,t} + h\,x_t$. Its variance is smallest at

$$h^* = -\frac{\mathrm{Cov}(r^{unh}_{SAA}, x)}{\mathrm{Var}(x)} = -\sum_i w_i \beta_i,$$

the negative of the SAA-weighted sum of the hedge ratios. The current policy corresponds to $h = -25.5\%$.

```
# unhedged excess return of the SAA
saa_unhedged <- drop(as.matrix(excess_unhedged[, names(asset_names)]) %*% saa_weight)

# variance-minimizing USD position, measured from the all-unhedged SAA
usd_position_opt <- -cov(saa_unhedged, currency_excess) / var(currency_excess)
stopifnot(all.equal(usd_position_opt, -sum(saa_weight * hedge_beta[, "beta"])))

# annualized volatility of the SAA for a given USD position h
saa_vol_with_usd <- function(h) sd(saa_unhedged + h * currency_excess) * sqrt(12)

usd_positions <- c(unhedged = 0, policy = -usd_hedge_saa, optimal = usd_position_opt)
usd_vols      <- sapply(usd_positions, saa_vol_with_usd)
```

Table 15: Volatility of the SAA for different USD positions, Jan 1999 to Sep 2026

SAA currency exposure	USD position	Volatility p.a.
All unhedged	0.0%	7.27%
Current policy	-25.5%	7.23%
Variance-minimizing	-14.3%	7.15%

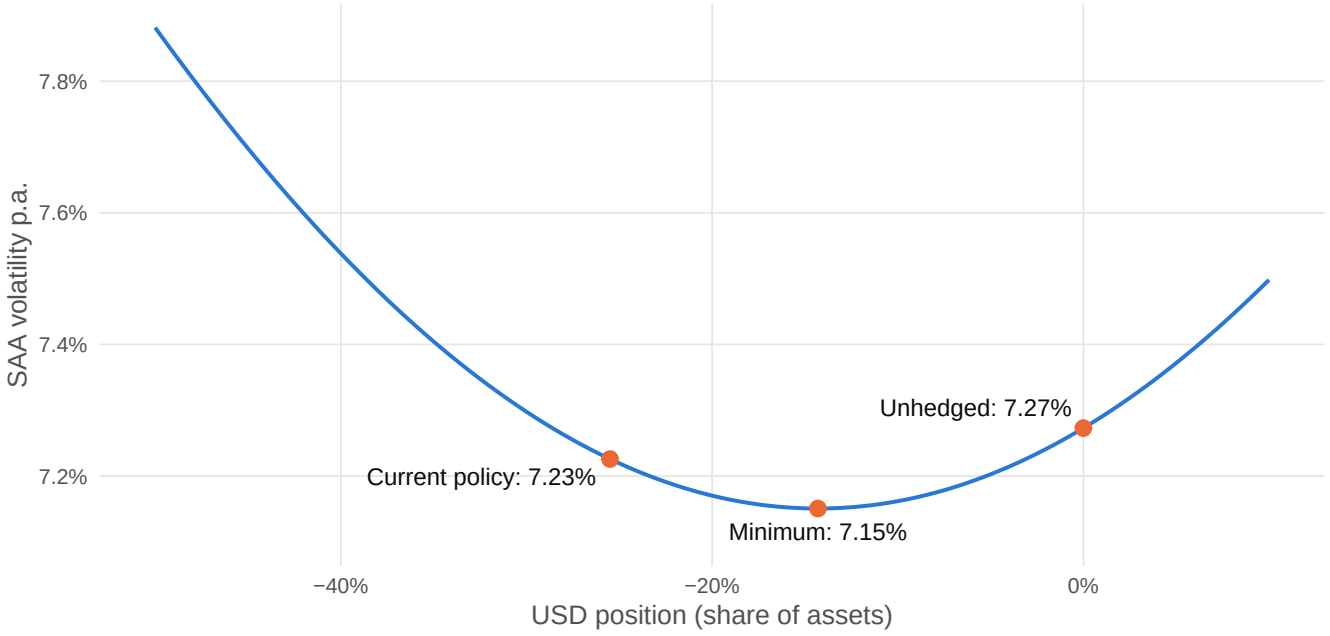


Figure 19: Volatility of the SAA as a function of its USD position (measured from the all-unhedged portfolio; negative = short USD)

The variance-minimizing USD position is -14.3% of assets, only a little more than half the current short of 25.5%. Most of it comes from US Treasuries and US equity; the other equity classes need little hedging and the euro assets even call for a small long dollar position. The SAA's volatility is 7.27% without any hedge and 7.15% with the variance-minimizing position (7.23% under the current policy). The whole range is only 0.12 percentage points, because the curve is flat around its minimum. For this SAA the currency hedge is therefore a minor risk decision, and the choice of policy should rest on cost and robustness (task 7d).

Task 7

```
# (a) is euro inflation + 2 pp attainable with the SAA over the remaining nine years?
horizon       <- 9
cash_now      <- panel$euribor3m[nrow(panel)] / 100   # expected euro cash rate: random walk from today's
↪  Euribor
inflation_exp <- panel$infl_5y5y[nrow(panel)] / 100   # market-implied euro inflation (5y5y swap)
target        <- inflation_exp + 0.02

# implied premia of the equity and bond blocks of the SAA (equilibrium of task 3)
bonds          <- setdiff(names(saa_weight), equity)
equity_premium <- sum(saa_weight[equity] * pi_eq[equity]) / sum(saa_weight[equity])
bond_premium   <- sum(saa_weight[bonds] * pi_eq[bonds]) / sum(saa_weight[bonds])

# glide path: equity share falls linearly from 45% to 25% over the nine years
equity_path  <- seq(0.45, 0.25, length.out = horizon)
premium_path <- equity_path * equity_premium + (1 - equity_path) * bond_premium

# expected compound return: cash plus average premium minus volatility drag (today's SAA volatility,
↪  conservative)
expected_compound <- cash_now + mean(premium_path) - 0.5 * saa_vol_full^2
shortfall_sd      <- saa_vol_full / sqrt(horizon)          # volatility of the nine-year average return
prob_target       <- pnorm((expected_compound - target) / shortfall_sd)

# same without the glide path (premium stays at 2.7 pp) as an upper bound
expected_no_glide <- cash_now + saa_premium - 0.5 * saa_vol_full^2
prob_no_glide     <- pnorm((expected_no_glide - target) / shortfall_sd)
```

(a) Is euro inflation plus 2 percentage points attainable?

The target is euro inflation plus 2 pp. With market-implied inflation of 2.13% (5y5y inflation swap), this is a nominal return of about 4.13% per year. The SAA earns the euro cash rate plus its premium. We assume Euribor stays at today's 2.62%. The board's premium of 2.7 pp applies to today's 45% equity; along the glide path to 25% equity it falls to 1.78%, using the implied premia of the equity block (5.24%) and the bond block (0.63%) from task 3. Over the nine years it averages 2.24%. After the volatility drag of 0.26%, the expected compound return is 4.59% per year, against the target of 4.13%.

The objective is therefore attainable in expectation, but with a thin margin. With a volatility of 7.23%, the nine-year average return has a standard deviation of about 2.41%, so the probability of reaching the target is only about 58% (65% without the glide path).

The target is in real terms, while 55% of the SAA is in nominal bonds, which lose when inflation surprises upwards, as in 2022. We therefore recommend presenting euro inflation plus 2 pp to savers as an ambition, not as an objective the product can be expected to meet.

```
# (b) risk of today's SAA and of the portfolio at the target date (25% equity, same mix within equities
↪  and bonds)
w_target <- c(saa_weight[equity] / sum(saa_weight[equity]) * 0.25,
```

```
              saa_weight[bonds]  / sum(saa_weight[bonds])  * 0.75)[names(saa_weight)]

year <- format(excess_policy$date, "%Y")
risk_profile <- function(w) {
  r   <- drop(excess_matrix %*% w)              # monthly euro excess return under the currency policy
  cum <- cumprod(1 + r)
  c(premium = sum(w * pi_eq),
    vol     = portfolio_vol(w, sigma_full),
    max_dd  = min(cum / cummax(cum) - 1),
    y2008   = prod(1 + r[year == "2008"]) - 1,
    y2022   = prod(1 + r[year == "2022"]) - 1)
}
risk_now    <- risk_profile(saa_weight)
risk_target <- risk_profile(w_target)
govt_2008   <- prod(1 + excess_policy$euro_govt[year == "2008"]) - 1
govt_2022   <- prod(1 + excess_policy$euro_govt[year == "2022"]) - 1
```

Table 16: Risk of the SAA today and at the end of the glide path, full-sample covariance, Jan 1999 to Sep 2026

	SAA today (45% equity)	At the target date (25% equity)
Expected excess return (equilibrium)	2.7%	1.8%
Volatility	7.2%	5.0%
Maximum drawdown since 1999	-30.0%	-18.2%
Excess return in 2008	-22.3%	-12.7%
Excess return in 2022	-15.8%	-15.9%

(b) Do the SAA and the pace of de-risking fit savers nine years from retirement?

A loss hurts more the closer it comes to retirement: the savings are largest then, there is little time to recover, and from 2035 savers withdraw capital or buy annuities, which locks a loss in. The same crash costs a saver far more in 2034 than in 2027, when new contributions still buy at low prices afterwards. Moving from 45% to 25% equity lowers the volatility from 7.2% to 5.0%, the largest drawdown since 1999 from 30.0% to 18.2%, and the 2008 loss from 22.3% to 12.7%.

The protection depends on what the bonds do in a crisis. In 2008 euro government bonds earned 4.3% above Euribor while equities collapsed. In 2022 they lost 18.6% together with equities, and the 25% equity portfolio lost 15.9%, against 15.8% for today's SAA. De-risking into nominal government bonds protects against recessions, not against inflation shocks, and task 2 showed that the stock-bond correlation has been positive since 2022 (in the euro area mostly since 2015). For savers who will buy annuities, government bonds still have a role: annuities become cheaper when yields rise, so losses on long bonds are partly offset by a lower annuity price. For savers who withdraw their capital there is no such offset.

We recommend keeping the pace of de-risking. Faster de-risking would lower the expected return further, and part (a) showed that the objective is already only narrowly attainable. Within the bond allocation, however, the product should not rely on euro government bonds alone as the safe asset, because they did not protect savers in 2022.

```
# (c) hedge fund index (Credit Suisse, net of fees): fully hedged euro excess return, months of task 1
hf_level <- panel$hf_hedgnav_usd[match(core$date, panel$date)]
hf_usd   <- simple_return(hf_level)
hf_r_eur <- (1 + hf_usd) * (1 + usd_return) - 1 - currency_excess - cash_return   # unhedged, minus full
↪  hedge, minus euro cash
common   <- !is.na(hf_r_eur)
```

```r
# monthly excess return of the SAA equity block (same mix as in the SAA)
equity_block <- drop(excess_matrix[, equity] %*% (saa_weight[equity] / sum(saa_weight[equity])))

# equity beta: contemporaneous, and summed with last month's equity return (stale pricing)
ok      <- which(common)[-1]
beta_hf <- coef(lm(hf_r_eur[ok] ~ equity_block[ok] + equity_block[ok - 1]))
hf_beta_now <- unname(beta_hf[2])
hf_beta_sum <- unname(beta_hf[2] + beta_hf[3])

hf_stats <- c(mean  = 12 * mean(hf_r_eur[common]), vol = vol_annual(hf_r_eur[common]),
              y2008 = prod(1 + hf_r_eur[common & year == "2008"]) - 1,
              y2022 = prod(1 + hf_r_eur[common & year == "2022"]) - 1)

# the board's proposal: 10 pp from equities into hedge funds, against the same 10 pp into the existing
↪  bonds
w_eq_cut <- saa_weight
w_eq_cut[equity] <- saa_weight[equity] * (0.35 / 0.45)   # equity from 45% to 35%, same mix within
↪  equities
w_alt <- w_eq_cut
w_alt[bonds] <- saa_weight[bonds] * (0.65 / 0.55)        # bonds from 55% to 65%, same mix within bonds
stopifnot(all.equal(sum(w_alt), 1))

path_stats <- function(r) {
  r <- r[common]
  cum <- cumprod(1 + r)
  c(vol = vol_annual(r), max_dd = min(cum / cummax(cum) - 1),
    y2008 = prod(1 + r[year[common] == "2008"]) - 1, y2022 = prod(1 + r[year[common] == "2022"]) - 1)
}
proposal <- sapply(list(saa         = drop(excess_matrix %*% saa_weight),
                        hedge_funds = drop(excess_matrix %*% w_eq_cut) + 0.10 * hf_r_eur,
                        bonds       = drop(excess_matrix %*% w_alt)), path_stats)

# return the hedge fund index would need to earn to be worth holding at the board's risk aversion
hf_required <- risk_aversion * 12 * cov(hf_r_eur[common], drop(excess_matrix %*% saa_weight)[common])
hf_period   <- format(range(excess_policy$date[common]), "%b %Y")
```

Table 17: The board's proposal against the same de-risking with bonds, euro excess returns, Jan 1999 to Sep 2026 (hedge funds: Credit Suisse Hedge Fund Index, net of fees, fully hedged)

	SAA	10 pp equity into hedge funds	10 pp equity into bonds
Volatility	7.2%	6.2%	6.0%
Maximum drawdown	-30.0%	-26.2%	-24.3%
Excess return in 2008	-22.3%	-19.8%	-17.6%
Excess return in 2022	-15.8%	-14.4%	-15.8%

(c) Should hedge funds replace part of the equity allocation?

We recommend against it. As a proxy we use the Credit Suisse Hedge Fund Index (net of fees, fully hedged into euro). The board's aim is to reduce risk, but hedge funds failed exactly when that mattered most: in 2008 the index lost 22.4% in excess of Euribor, as much as the SAA itself (22.3%). Its equity beta of 0.25 rises to 0.31 once last month's equity return is included, a sign that reported returns are smoothed. Moving the same 10 pp from equities into the existing bonds lowers the risk of the SAA more: volatility 6.04% against 6.21%, maximum drawdown 24.3% against 26.2%, and a 2008 loss of 17.6% against 19.8%. Hedge funds did better in 2022 (1.1% loss), when stocks and bonds fell together, which is their main argument.

The index earned 3.8% per year above Euribor, more than the 1.35% it would need to be worth holding at

the board's risk aversion. On returns alone, hedge funds would therefore have been attractive. But this overstates what the product could expect, because index returns are biased upwards: funds that close or stop reporting drop out of the index. More importantly, the board's aim is to reduce risk, and for that the existing bonds are the better tool. Hedge funds also bring high fees, lock-ups and the possibility of suspended redemptions, which fit poorly with a retail pension product that has outflows from 2035. Given part (b), we do not see a need to de-risk faster than the glide path; if the board wants to, the existing bond allocation does it more cheaply and with less risk.

```
# (d) proposed policy: hedge US Treasuries fully and US equity by half, stop hedging the other foreign
↪  equities
hedge_ratio_new <- hedge_ratio
hedge_ratio_new[c("dev_equity", "em_equity")] <- 0
usd_hedge_new <- sum(saa_weight * hedge_ratio_new)

# USD positions measured from the all-unhedged SAA (negative = short USD)
policy_options <- c(unhedged = 0, current = -usd_hedge_saa, proposed = -usd_hedge_new, min_variance =
↪  usd_position_opt)
policy_vol     <- sapply(policy_options, saa_vol_with_usd)
policy_cost    <- abs(policy_options) * expected_x      # expected carry lost per year by the short USD
↪  position

# stability check: hedge ratios over the last five years
last_60 <- tail(seq_along(currency_excess), 60)
beta_5y <- sapply(usd_assets, function(a) unname(coef(lm(excess_unhedged[[a]][last_60] ~
↪  currency_excess[last_60]))[2]))
```

Table 18: Currency policy options for the SAA: volatility (Jan 1999 to Sep 2026) and expected cost of the dollar hedge at today's interest rate differential

Currency policy	USD position	SAA volatility	Expected hedging cost p.a.
All unhedged	0.0%	7.27%	0.00%
Current policy	-25.5%	7.23%	0.44%
Proposed policy	-19.0%	7.16%	0.33%
Variance-minimizing	-14.3%	7.15%	0.25%

(d) Should the currency policy be changed?

We recommend a small change: keep the full hedge of US Treasuries and the half hedge of US equity, but stop hedging developed ex US ex euro area and emerging markets equity with dollar forwards. Task 6 showed that the hedge ratios that minimize risk are close to the policy for US Treasuries (0.90) and US equity (0.52), but far below one half for developed ex US ex euro area (0.21) and emerging markets equity (0.03). Their currency risk is in yen, pounds and emerging market currencies, not in the dollar, so a dollar hedge adds risk instead of removing it. Over the last five years their hedge ratios were even negative (-0.28 and -0.37), so the conclusion does not depend on the early part of the sample.

The proposed policy reduces the dollar short from 25.5% to 19.0% of assets, closer to the variance-minimizing 14.3%, and lowers the SAA volatility from 7.23% to 7.16%. The risk difference between all options is small, so the main gain is cost: with the dollar following a random walk, the short dollar position loses the interest rate differential of 1.73% per year, and the proposed policy saves 0.11% of expected return per year, which helps with the objective in part (a). Over the long run this saving should disappear if uncovered interest parity holds, so it is not the reason for the change, only a benefit of it.

We do not recommend hedging exactly at the variance-minimizing ratios. They are estimated with error and change over time, and a rule based on which currency an asset is actually exposed to is easier to explain to savers and more robust than one based on regression estimates.